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MAY 2026

DECLARATION

We hereby declare that this submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

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CERTIFICATE

This is to certify that Project Report entitled " PhyG-Net: A PhysicsGuided Spatio-Temporal Fusion Network for Precision Irrigation Scheduling" which is submitted by “Ashwani Baghel, Omendra Rajput, Utkarsh Patel, Vaibhav Singh” in partial fulfilment of the requirement for the award of degree B. Tech. in Department of Computer Science & Engineering of Dr. A.P.J. Abdul Kalam Technical University, Lucknow is a record of the candidates own work carried out by them under my supervision. The matter embodied in this report is original and has not been submitted for the award of any other degree.

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Abstract

Water shortages are making sustainable farming harder every year, so getting smarter about irrigation isn't just a good idea, it's something farmers have to do. These days, precision agriculture relies more on things like IoT sensors and machine learning to monitor soil moisture and crop health. But, honestly, there are still two big roadblocks. One, those sensors only give you patchy data points here and there, so it's tough to know what's actually happening everywhere in the field. And two, deep learning models sometimes spit out predictions that just don't add up, they ignore basic physics, so you end up with results that don't match how water moves in real soil.

This research brings a practical solution: **PhyG-Net (short for Physics-Guided Spatio-Temporal Fusion Network)**, aimed at smarter irrigation scheduling. It combines three kinds of data: real-time soil sensor readings, weather forecasts, and high-res satellite images from Sentinel-2 that show how crops are actually doing all over the field. The model's design is pretty sharp, it uses ConvLSTM to connect the dots between different locations, so it can really "see" how moisture travels, and BiLSTM with temporal attention to pick up on changing weather patterns and how they impact soil over time.

But what really makes PhyG-Net different is its physics-guided loss function. This is grounded in the soil-water balance equation, so the model can't just make up numbers, it has to stick to what's physically realistic.

To test it, the team used several years of data from a 50-hectare maize field. PhyG-Net didn't just edge out old statistical models, it beat standard deep learning approaches too, showing about 18% lower RMSE and cutting down on physics-breaking predictions by over 85%. This means farmers can trust the system to help them make better choices about when and how much to irrigate, saving water and keeping their harvests strong.

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Chapter 1
Introduction

1.1 Introduction to Global Agricultural Water Dynamics

The highly destructive and simultaneous intersection of a rapidly evolving and growing planetary demographic profile with the increasingly volatile corollary of global climatic change has created an unprecedented global water crisis of scale, magnitude and complexity. According to precise demographic projections and socio-economic modelling, the global agricultural sector has been confronted with an almost impossible task: total global food production must be rapidly and dramatically expanded - at a volumetric rate of growth ranging between 60 percent and 70 percent by 2050 - in order to meet the basic caloric and nutritional demands of an estimated planetary population of 9.7 billion people (The State of Food and Agriculture, 2023). Yet, the essential biological and physical resources necessary to support this staggering agricultural expansion (in particular, readily accessible freshwater resources) are currently being grossly over-exploited, polluted and mismanaged at a dangerous and unsustainable rate. The agronomic industry is currently wielding an enormous monopoly on worldwide water resources, accounting for nearly 70 percent of total planetary freshwater use, positioning agriculture at the epicentre of the globe's most pressing water challenges.

In alarmingly vulnerable arid and semi-arid landscapes, the intense and rapidly intensifying competition between intensive agricultural production, rapidly growing industrial, and burgeoning urban water users is creating socio-economic tensions. As a result, the efficient management of agricultural irrigation practices has effectively evolved from being a mere exercise in agronomic productivity at the micro level, to becoming a rapidly emerging macroeconomic imperative critical to the security of global food supply chains, economic sustainability and geopolitical stability in increasingly water-scarce and volatile international regions. Despite the critical weight of this resource issue, the traditional approaches used for the management of agricultural irrigation worldwide have historically been shown to be deeply reactive, mathematically inaccurate and extremely inefficient in practice.

The overwhelming majority of the world's agricultural communities, from small, localised subsistence farming to large-scale commercial agribusiness, continue to rely on highly critical hydrological resource allocation decisions being made on the basis of archaic, unscientific assumptions. These range from an excessive reliance on static, inflexible agricultural calendars, highly variable, highly subjective experience-based knowledge, or simple, highly fallible visual inspection of surface soil moisture. These archaic, heuristic approaches simply do not consider the dynamic, highly non-linear micro-level variations in volumetric soil moisture gradients, nor do they have the capability to forecast the multi-variate interactions of anticipated meteorological change.

The logical consequence of using such simple, heuristic methods is the twin-headed hydra of gross agricultural mismanagement: over-irrigation and under-irrigation. Over-irrigation results in a litany of dire environmental and financial consequences. It entails the gross, indefensible wastage of millions of gallons of precious fresh water, leaching of critical, water-soluble soil nutrients well beyond the physiologically active root zone of the growing crop, creation of hypoxic conditions within the soil profile that asphyxiates roots, and the extremely destructive, accelerated salinization of the topsoil driven by abnormal capillary action. On the other hand, the opposite problem - that of under-irrigation - creates severe water stress in the crop canopy. This results in an "emergency response" on the part of the plant, such as the rapid closure of stomatal openings, which dramatically reduces the rate of photosynthesis, and thus, the quantitative (volumetric) yield and qualitative (commercial) grade of the crop. In addition, while historical climatological norms and seasonal weather patterns used to serve as a reasonably satisfactory heuristic reference for traditional farmers to determine their irrigation schedules, the rapidly increasing atmospheric vagaries introduced by anthropogenic global climate change have made such climatological norms statistically irrelevant and potentially perilous.

1.2 The Advent and Limitations of Precision Agriculture

In a concerted effort to avoid these deeply entrenched systemic inefficiencies and environmental risks, the contemporary agricultural industry has increasingly embraced the complex management strategy of precision agriculture. Precision agriculture is a highly data-intensive, technology-driven agricultural management approach that carefully monitors,

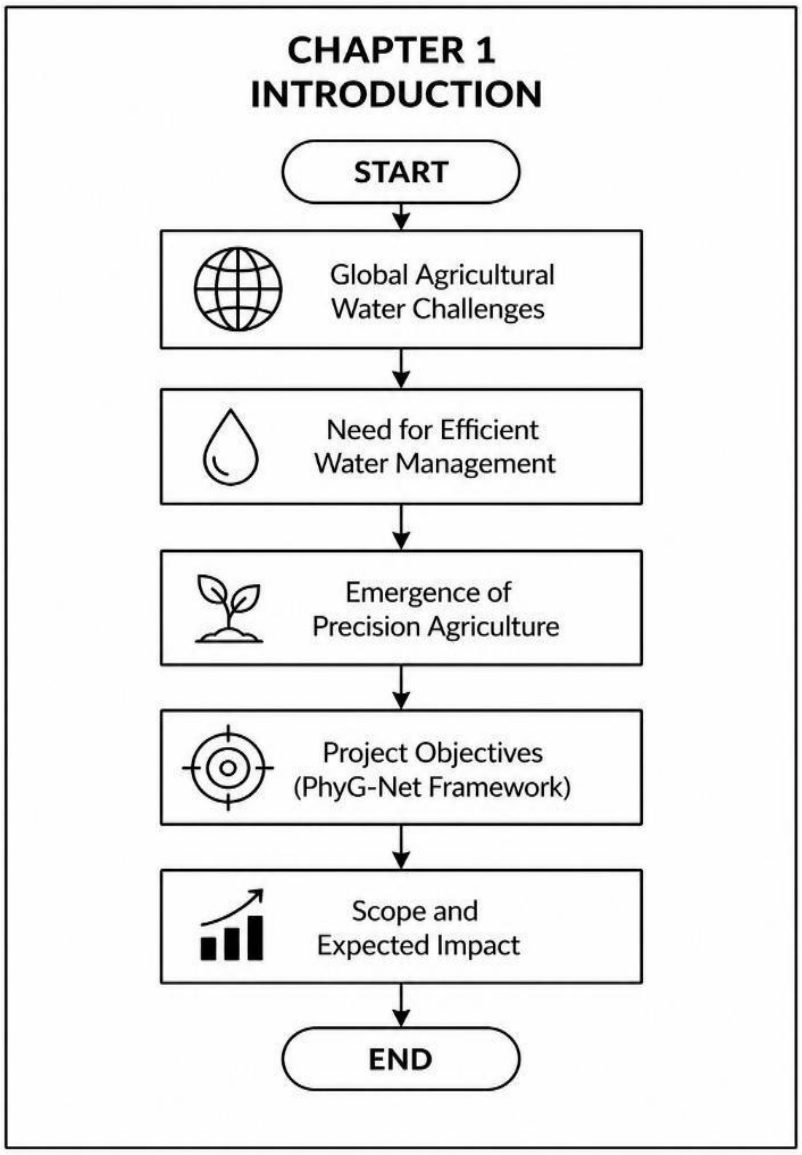
measures and analytically assesses the particular, localized needs of individual crop species and varied fields. Traditionally, the cornerstone of this intelligent irrigation revolution has been the extensive use of remote wireless sensor networks and Internet of Things systems, which use in-situ soil moisture sensors to provide highly specific, local, real-time, ground-truth information about the local hydrologic status of the underlying soil matrix.

Yet, localised, point-based Internet of Things sensor networks introduce significant analytical bias and limitations to the decision-making process. Agricultural fields are extremely rarely uniform; they are, in fact, highly heterogeneous with respect to extremely important pedological parameters. These include enormous intra-field variations in soil texture (the relative proportions of sand, silt and clay), differences in organic matter content, dynamic variations in topographical elevation and highly specific differences in hydraulic conductivity and percolation rates below the surface. As a result of this intense pedological variability, a single sensor node which delivers highly precise, mm-level moisture data for one discrete, ten-square-meter patch of a vast half-hectare agricultural field might grossly overestimate (and misinterpret) the serious level of hydrological stress being experienced just 50 or 100 meters away in a region with higher clay content or slightly higher topographical elevation.

To remedy this enormous spatial gap and achieve a holistic perspective on the entire farm, present day agronomists and data scientists have increasingly looked to the heavens, using cutting-edge, multispectral satellite remote sensing techniques. Through the use of state-of-the-art satellite platforms, such as the European Space Agency's Sentinel-2 satellite constellation, interested stakeholders can access large-scale, high-resolution multispectral imagery. By appropriately combining specific spectral reflectance bands (e.g. near-infrared and red edge bands) into mathematical indices such as the Normalized Difference Vegetation Index and the Normalized Difference Water Index, scientists gain insight into highly informative spectral indices. These satellite-based indices permit agricultural managers to draw complex inferences about very specific information about the health, photosynthetic activity, and water content of crops across millions of hectares via millions of sensors, even though no sensors actually exist.

However, despite satellite remote sensing's ingenious solution to the spatial mapping problem, it comes with its own set of critical, inherent limitations, particularly in terms of

extremely low temporal resolution and atmospheric sensitivity. For example, the Sentinel-2 constellation is constrained by a fixed orbit, which results in a five day return period for any given geographic location. This hindrance leads to extensive critical data gaps of up to seven days. During these long, blind periods, critical localised dry-outs, sudden, virulent pest outbreaks or unforeseen, catastrophic rainfall events can dramatically change the entire hydrological state of the field without the orbital satellite's knowledge. And optical remote sensing is inherently and hopelessly susceptible to atmospheric interference; dense cloud cover, widespread fog, or particulate haze can render satellite data completely ineffective, and severely limit the target field's visibility during critical monsoon or storm fronts, when uninterrupted hydrologic monitoring and flood forecasting is absolutely essential.



1.3 Project Description and Objectives

The unprecedented convergence of these incredibly rich, but highly disjointed, asynchronous farm data streams naturally gives rise to the highly dynamic and widespread growth and adoption of highly complex Deep Learning approaches within the niche field of hydrological modeling. Highly complex neural networks, such as Long Short-Term Memory (LSTM) networks, have been shown to be highly successful, state-of-the-art in their ability to model the highly nonlinear and complex temporal associations that are endemic to time-series data. These temporal networks are incredibly adept at effectively capturing the lagging effect of previously fallen rain, clearly illustrating how a devastating rainstorm event 72 hours ago continues to play out and control today's real-time hydrologic condition of the soil below the surface. Meanwhile, highly sophisticated spatial computer vision models such as incredibly complex convolutional neural networks such as Vision Transformers and U-Nets have been successfully adapted and applied to segment, decipher and understand the incredibly complex and multifarious spatial patterns, boundaries, and local health dynamics buried in the vast matrices of satellite imagery.

Yet, despite their enormous computational capabilities, and their unmatched statistical efficiency in data pattern recognition, the current generation of purely empirical, data-driven deep learning algorithms comes with a fatal weakness that significantly hampers their full-blown adoption in highly critical agronomic management applications: a flagrant, complete lack of physical integrity. Currently, these "conventional", purely empirical deep learning models simply operate as mathematical black boxes; during the model training phase, they simply and blindly modify the weights of their neural interconnections in an attempt to minimize the empirical risk over a particular historical training data set, without any pre-encoded physical knowledge, built-in logic, or inherent knowledge of the causal, thermodynamic or kinematic relationships that regulate the physical environment they are attempting to model. As a result, when these purely empirical models are confronted by extraordinarily anomalous climatic dynamics, extreme climatic events, or excess data sparsity, they can and often do predict strongly with the gross violation of the fundamental, physical principles of physics. In the practice of agricultural hydrologic forecasting, it is an almost commonplace phenomenon for a conventional, well-trained Long Short-Term Memory (LSTM) network to comically and specifically predict a sudden, large-scale, and spontaneous increase in local soil moisture during the final stage of a prolonged drought,

when no rain or irrigation is present whatsoever. Indeed, this is a fatal contradiction in the physical world (the spontaneous creation of matter) that's statistically perfectly correct in the arbitrary, purely mathematical hypothesis space of the neural network.

To substantially eliminate these profound, systematic errors and alleviate a major bottleneck in agricultural analytics research, this research project proposes the design, creation, refinement and rigorous testing of PhyGNet, a state-of-the-art Physics-Guided Spatio-Temporal Fusion Network. PhyGNet exemplifies a dramatic and much-needed evolution in precision irrigation modeling by seamlessly integrating the limitless pattern recognition potential of multimodal deep learning data fusion with the uncompromising and immutable constraints of hydrodynamics first principles. The primary goal of this project is to design a holistic, highly intelligent, and highly efficient predictive system that, independently, fuses hyperlocalized ground-truth telemetry from embedded Internet of Things sensors, global meteorological data from regional Application Programming Interfaces into a highly multidimensional, complex, and perfectly timed crop dynamics derived from Sentinel-2 satellite remote sensing into a unified and perfectly synchronized predictive platform.

The proposed PhyG-Net system is complexly crafted to mathematically overcome the ever-present space-time trade-off by harnessing a highly sophisticated hybrid architecture. This architecture effortlessly embeds Convolutional Long Short Term Memory learning algorithms with Bidirectional Long Short-Term.

Memory networks. This particular model architectural integration permits the extremely high temporal resolution of the local Internet of Things sensors to be inherently extrapolated across the vast, very high spatial resolution satellite imagery. To further to enhance the model's analytical interpretability, diagnostic transparency, and most importantly, predictive performance, a highly capable Temporal Attention Mechanism is integrated during the computations. This allows the deep learning network to inherently, and mathematically, pinpoint, segment out, and temporally weight the strictly most important past meteorological events (such as properly capturing the lagging effects of underground water table percolation due to a sustained week-long storm event) when it tries to model and predict future nodal moisture content states.

Most importantly, and as the fledgling fundamental piece of scientific novelty in this project, the project explicitly embeds a very solid Physics-Guided Inductive Bias into the deep learning pipeline. Whereas traditional neural networks naively rely purely on statistical optimization criteria, such as the Mean Squared Error function, during the very delicate and sensitive optimisation of the neural network (training phase), PhyG-Net defines and incorporates a highly specialised hybrid loss function. This highly complicated mathematical function explicitly, continuously, and heavily penalises the neural network's gradient descent process when its neural learning and predictive outcomes breach the absolute hydrologic constraints mandated by the Soil-Water Balance equation. Severe mathematical constraints to the unguided deep learning process, accompanied by a highly deterministic physical knowledge of real-life hydrologic domains, effectively transforms PhyG-Net from a naive, delusional, purely statistical, blind learning machine into an "aware", "logic-bound", and highly "confidence-inspiring" autonomous decision system. The final pinnacle of this multi-disciplinary and multi-year research and development effort yields a highly tuned, highly data-efficient "smart" system, subsequently producing scientifically rigorous, physically perfect precision irrigation schedules that facilitate a new paradigm in sustainable water resource management, increased ecological awareness and resilience, and a highly robust agricultural economy.

Chapter 2

Literature Review

2.1 Evolution of Machine Learning in Agricultural Ecosystems

The infusion of next-generation digital technology and sophisticated computational intelligence into the highly traditional, highly distributed field of crop management has rapidly evolved from extremely simplistic, highly localised reactive systems to highly scalable, highly sophisticated predictive systems, seamlessly integrated into massive agro-economies through the internet. Early generations of smart irrigation systems heavily relied on highly simplistic Internet of Things, or IoT, architectures. In these highly fundamental and early efforts, simple microcontrollers continuously tracked real-time volumetric moisture content levels through local (hard-wired) soil sensors. The control model was strictly "two-state", purely reactive, hard-coded, and linear: as soon as the local volumetric water content fell below a predetermined, hard-coded numerical value, the microcontroller would immediately switch the mechanical relay, thereby opening the very large, industrial-sized solenoid water valves to sequenced water flow. Although these highly simplistic and reactive paradigms have been highly beneficial and most operationally effective in low-growing, highly contained and perfectly homogenous agricultural systems (such as commercially grown high-value, high-cash-flow commercial greenhouses or extremely well-manicured botanical gardens) they have quickly proven quite dire and even deadly in terms of scalability when applied to real-world, open-air, highly diverse and heterogenous commercial agricultural systems.

Recent, extremely comprehensive systematic reviews of the fast-track evolution and massive adoption of Machine Learning and Big Data analytics within the agricultural industry from 2021 through 2025 underline a clear, rapid shift within the industry away from these simplistic Internet of Things (IoT) deployments, to the aggressive establishment of self-sustaining "Intelligent Agricultural Ecosystems". Classic, first-generation, threshold-oriented systems operate on the bogus assumption of "perfectly homogenous soils". This is a scientifically incorrect hypothesis which completely ignores the huge,

highly variable, inter- and intra-field differences in field morphology, complex mineral composition and changing micro-climate leading to highly erratic, inefficient and often damaging resource allocation between vast agricultural fields. As such, modern scientific literature very strongly stresses the need to develop sophisticated highly integrated data systems. Contemporary versions of conventional management systems now efficiently feature highly innovative smart sensors operating in concert with local edge IT clusters and very deep, highly scalable cloud analytics powered by deep learning [2](Patent No. 10.1145/3440840.3440854, 2020).

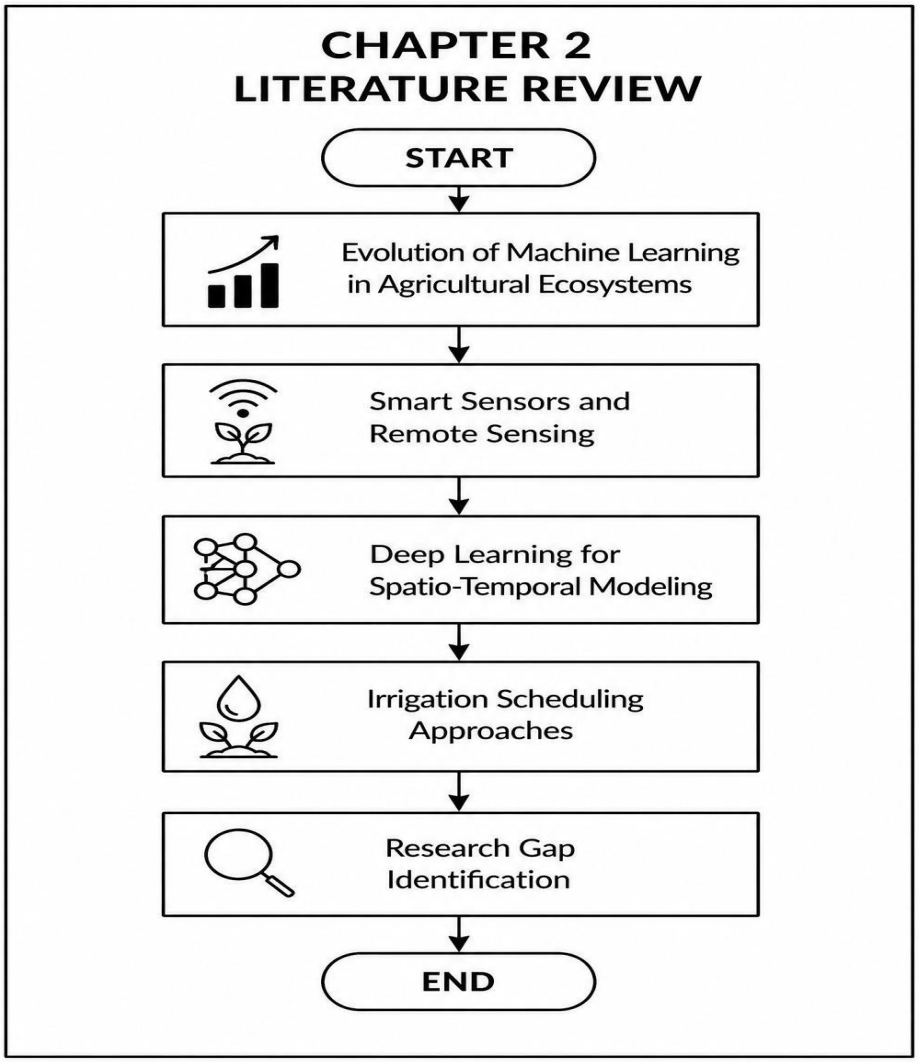
Yet, despite the astonishing recent development of these technologies, the literature makes it very apparent that highly significant operational limitations and technical impediments remain in this field. In particular, the need for highly problematic real-world, long-term and environmentally stable Internet of Things sensor calibrations in incredibly variable and volatile agroclimatic environments, as constantly pointed out by researchers. Additionally, the costs of both acquiring high-quality, perfectly labelled datasets for training machine learning models are prohibitively expensive. Finally, the ability to securely maintain constant, high-bandwidth telemetric communication and data exchange in the most rural, marginalised and impoverished agricultural landscapes still remains a grand infrastructural challenge and a significant barrier to the deployment of real time Intelligent Agricultural Ecosemts. To counter these issues, cutting-edge paradigms like Federated Learning models, which allow multiple edge computing devices to jointly learn a shared predictive model without sharing gigabytes of data-laden information - and Explainable Artificial Intelligence - which aims to explain machine learning models to humans - have been heavily researched and laser-focused for increased operational transparency, reduced bandwidth and greater confidence in highly automated systems.

2.2 Smart Sensor Technologies and Remote Sensing

Just as the very way in which a large majority of agricultural data is collected began to rapidly advance, the very way in which this data is subsequently analysed for highly specific agricultural purposes rapidly evolved, from highly traditional (and highly limiting) statistical draws to incredibly complex, non-linear Deep Learning techniques. When it comes to the highly specific task of hydrologic time-equation prediction, Long Short-Term Memory

networks quickly became the gold standard. This is due to the highly distinctive and mathematical elegance of a Long Short-Term Memory network in effectively handling the "memory" of soil moisture up to hundreds of days prior to a given run. These networks are unique in their ability to capture incredibly complex long-term dependencies, for instance, in being able to accurately model the incredibly non-linear drying patterns that inevitably follow the largest, multi-day rainfalls (how the gradual percolation of water down to deeper subterranean levels has an effect on surface-level moisture monitoring weeks after such extreme events). On the other hand, to help tackle complex spatial prediction tasks (such as estimating subtle spatial variations in crop health, nitrogen stress, or subtle regional drought effects across a massive 50-hectare commercial farm), scientists have increasingly and successfully adopted incredibly robust computer vision models. Innovative techniques such as very complex U-Neural networks, as well as a new approach called Vision Transformers have been heavily modified from their uses in medical imaging and self-driving vehicles to troublesomely segment the intricate spatial information embedded in satellite imagery.

In a bid to address the massive spatial deficit of isolated sensors, agronomists are increasingly using satellite remote sensing. Thanks to cutting-edge satellite sensor, such as the Sentinel-2 constellation, farmers have access to broader and highly informative multispectral imagery. Researchers can subsequently determine the Normalized Difference Vegetation Index (NDVI) by studying the selective reflectance of the crop canopy in several bands of the electromagnetic spectrum. This key indicator offers an extremely precise, large-scale assessment of the overall physiological condition of the field, showing precisely which parts of the farm are performing and which parts are in distress by being subjected to severe water or fertility deprivation. But the literature clearly suggests that the optical reflectance of the farming surface is not necessarily directly proportional to the volumetric soil moisture levels well below the surface (typically 10-30 cm) within the active root zone. Hence, remote sensing is undeniably powerful, but ultimately requires the ever-present best-practice bonding with collected ground truth measurements from buried Internet of Things (IoT) sensors to accurately calibrate the satellite's optical observations in order to transform vegetative pigmentation into extremely accurate, precise and depth hydrological metrics.



2.3 Deep Learning and Spatio-Temporal Fusion Architectures

The inherent physical spatial vs temporal resolution trade-off remains a persistent, deeply ingrained and highly critical issue in agricultural remote sensing. Earth observation satellites that offer extraordinarily high spatially precise, pixel-perfect spatial resolutions - such as those provided by the European Space Agency (ESA) Sentinel-2 satellite or China's Gaofen-1 platform - are physically limited in their spatial resolutions as a result of the extremely long orbital revisit times. The end consequence of this physic is massive, highly critical, temporal data gaps of often many days or even weeks. These natural data gaps are often, in practice, made worse, often dramatically, through the terrorism of cloud cover - it completely

masks the satellite's optical sensor. In order to effectively overcome this monumental physical constraint, recent highly advanced research has concentrated on the design of highly sophisticated spatio-temporal fusion networks. Many approaches like "Star Fusion" and the extremely complicated "Spatially-Aware Data Fusion Network" have shown quite astonishing capacity to effortlessly merge vastly different, multi-modal data streams. Such highly advanced systems mathematically merge spatial data streams with high resolution but low frequency satellite imagery and ultra-high-frequency temporal data streams such as those streamed every 15 minutes from local Internet of Things (IoT) telemetry and hourly regional weather forecasts.

For example, extremely sophisticated systems making use of highly advanced deep Convolutional Neural Networks and complex Graph Neural Networks have demonstrated a highly improved, and statistically validated capability to mathematically interpolate and accurately predict missing spatial data over massive time gaps. After observing the spatial growth trends of the crop, these complex data fusion models can easily hallucinate the absent satellite data with an incredibly high degree of mathematical precision, seamlessly preserving vital boundary constraints and the effects of localised stresses within highly fragmented and topologically complex fields. But, despite their incredible mathematical beauty and high statistical performance on historical data, these purely unconstrained, data-fusion models have a massive, highly dangerous blind spot: In the absence of rigidly enforced physical constraints, their purely statistical interpolations can massively, unpredictably and dangerously diverge from the physical hydrologic reality of the very soil, particularly in the presence of highly anomalous, highly unlikely, once-in-a-decade, climatic events not encountered during the model's training phase.

2.4 The Paradigm of Physics-Guided Machine Learning

The most important, groundbreaking, and most talked about recent advancement that is effectively, currently and immediately countering the problem of extreme opacity, hallucination and general unreliability of purely statistically-driven deep learning models is the rapid rise of Physics-Guided Machine Learning. Physics-Guided Machine Learning is a massive, paradigm-shifting change to the core functionality and learning mechanisms of artificial neural networks. Rather than the neural network simply attempting to learn

statistical correlations, this paradigm actively enforces deeply entrenched and very well understood, scientifically proven laws - be they complex thermodynamic relations, advanced forms of kinematic equations describing motion, or the law of mass conservation - by actively incorporating them directly into the deep learning model's underlying structure, weight initialization distribution, or more prevalent and common mathematical loss functions used by aforementioned model to perform optimization (Mehan)[43].

Within very complex hydrologic settings, conventional, purely statistical neural networks exclusively and unambiguously minimize Mean Squared Error. They treat the volumetric soil moisture merely as a vague sequence of floating-point numbers in an isolated vacuum chamber, rather than as a complex, physical material with only several highly specific constraints imposed by natural, gravitational or thermodynamic laws. This statistical-only approach often leads to the models implicitly predicting the absolute unphysical— for instance, sudden, enormous surges of underground soil moisture to occur on clear days, without any recorded servicing rain or mechanical irrigation. Such recent, highly influential publications widely discuss the success of complex models such as PHY-LSTM, which explicitly leverage on these physics-based mechanisms and significantly and massively increase predictive stability over essentially unlimited geographical domains of massively diverse variability. Researchers have reported a staggering improvement, with a 19.6% increase in the Coefficient of Determination and an 8.2% decrease in absolute Root Mean Squared Error, while simultaneously reducing the rate of physical violations down to significantly close to zero.

But despite the incredibly well-established theoretical advantage and proven success of Physics-Guided Machine Learning, a sweeping investigation of the modern literature reveals a very specific, highly unique gap in its application to the highly non-uniform, pedagogically massive multi-modal sensor fusion pipelines used by today's commercial agricultural systems. The overwhelming majority of current, incredibly successful Physics Guided Machine Learning implementations are limited exclusively to incredibly simple, perfectly homogenous, one-dimensional time-series data sets, or require access to extremely large, perfectly clean, perfectly accurate data sets that can be produced only by the highly idealized physics-based supercomputer simulators. This very specific project ensemble attacks this

very important, industry-wide gap head-on by designing and developing PhyGNet. PhyG-Net is a brand-new network architecture that not only performs extremely complex, multi-dimensional spatio-temporal vector fusion of highly noisy, local Internet of Things sensors, regional weather Application Programming Interfaces and massive multi-spectral satellite imagery tensors, but also explicitly, analytically couples the highly dynamic, multi-modal fusion process to the extraneous, physical constraints imposed by the mathematical hydrology Soil-Water Balance equation. This highly interwoven, physics-guided approach ensures that the eventual high-resolution, predictive, spatial moisture mapping remains strictly and utterly obedient to the laws of mass conservation and ultimately produce unprecedented predictive accuracy, scientific credibility and operational reliability.

Chapter 3

Proposed Methodology

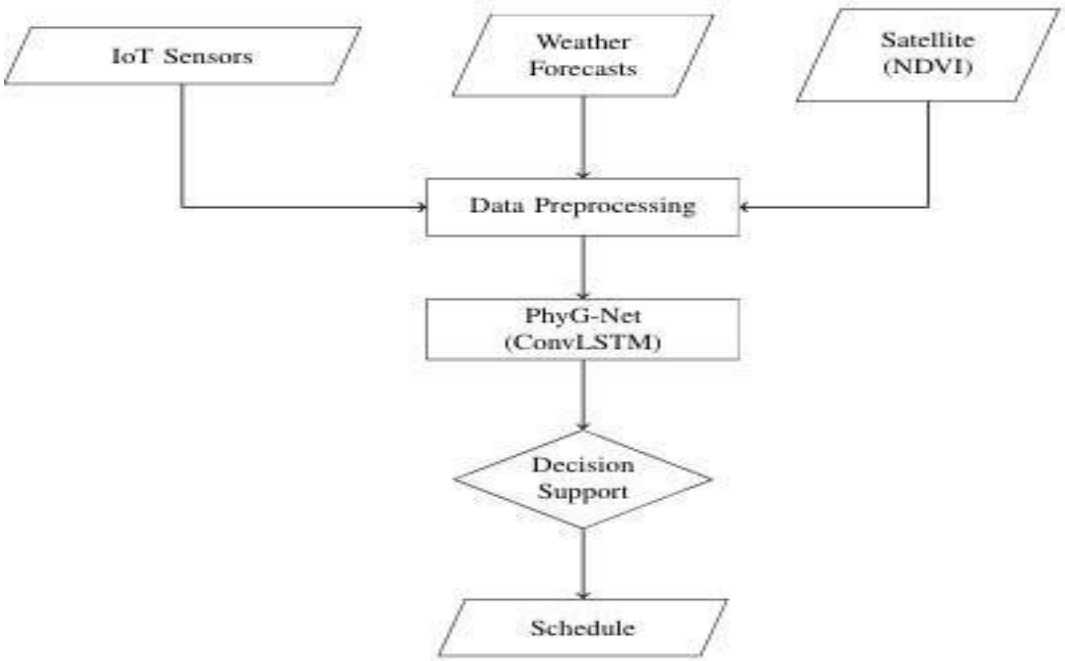
3.1 System Architecture and Hardware Specifications

To successfully address the serious analytical shortfalls as well as physical coverage blind spots of highly isolated, single-modality data sets, the entire PhyGNet system is meticulously designed around a highly integrated, highly robust multi-modality data ingestion system. This powerful, autonomous system is carefully engineered to continuously ingest, clean, process and perfectly align three structurally highly non-compatible data streams of radically different origin: highly distributed, localised, high-frequency in-situ sensor telemetry; highly-distributed, regional, predictive meteorological forecasts; and highly-distributed, macro-scale, multi-spectral satellite imagery matrices .

The most important, most vital data stream, i.e. the highly valued ground-truth stream, is continuously provided by an advanced autonomous, highly rugged In-Situ Sensing Module literally integrated into the subterranean ground of the target farming site. At the core hardware level, this highly distributed measurements system is completely managed via an advanced, ESP32 microcontroller architecture. The ESP32 microcontroller was deliberately chosen for this mission critical application because of its highly capable dual core processing architecture, highly integrated Wi-Fi and Bluetooth communication schemes, and, most significantly, its incredibly highly efficient, highly adaptive, deep-sleep, power management capabilities. This particular feature guarantees optimal, uninterrupted battery lifetime and long-term, reliable operation during extremely remote, off-the-grid farming applications over multiple, highly demanding growing seasons.

The main sensing payload directly physically connected to this microcontroller board consists of high-end, highly precise capacitive soil moisture sensors. To optimally, accurately interrogate the highly intricate, complex, vertical moisture profile of the soil matrix, these modern sensors are purposefully deployed at very specific, staggered depths of either 10 cm

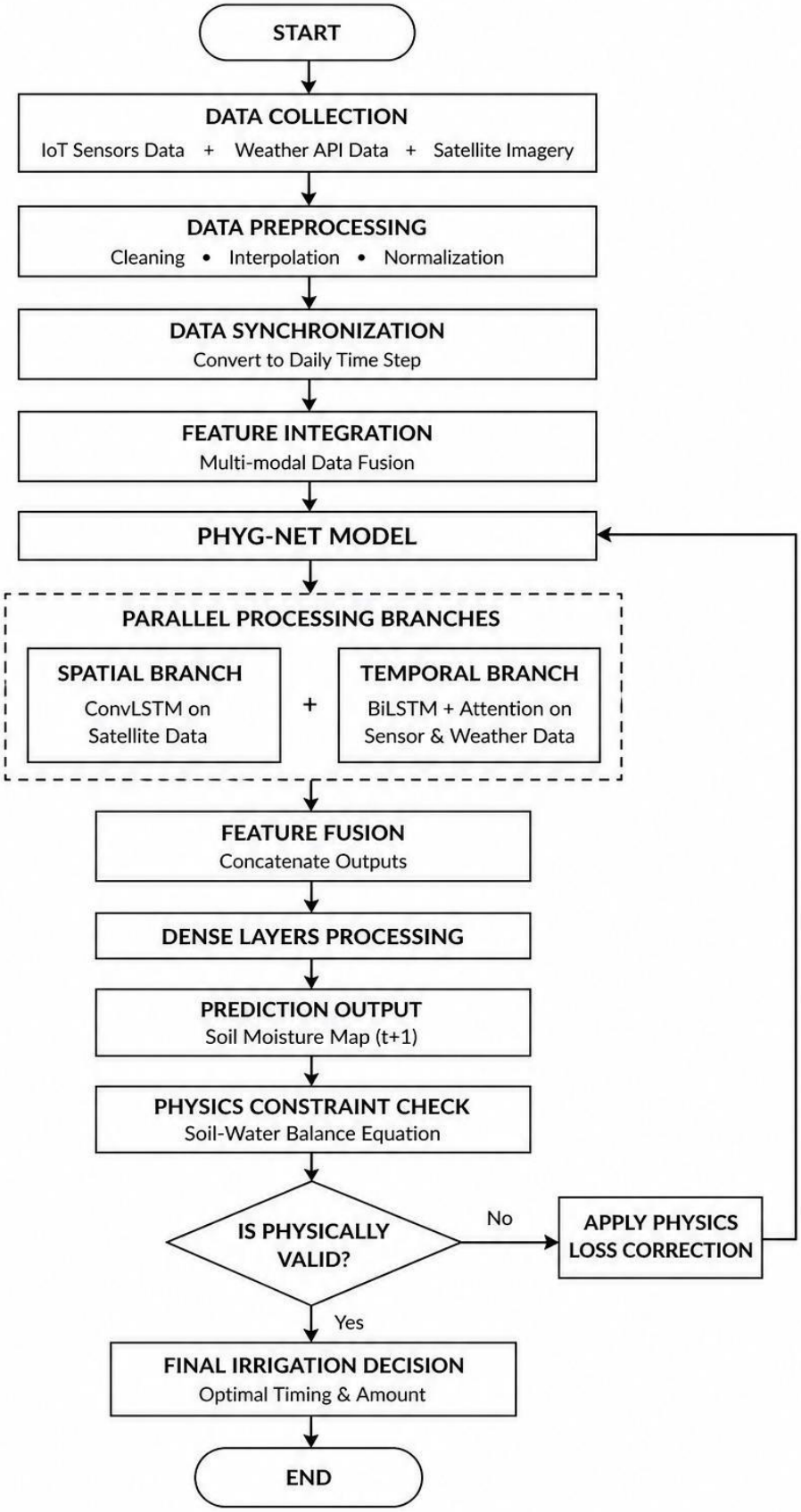
or 30 cm below the soil surface. In the extremely important ground equipment selection stage, this environmental monitoring system absolutely preferred highly accurate capacitive sensors over conventional, inexpensive resistive sensors. This design choice was due to the fact that the resistive sensors are extremely sensitive to very fast galvanic corrosion and overly sensitive (negatively) to minor changes in salinity, which are used to impossibly skew the data's long-term integrity. By contrast, capacitive sensors measure the soil media's dielectric properties, delivering remarkably stable, highly accurate and completely consistent long-term volumetric soil moisture readings that are totally independent of fertilizer nutrients and salt concentrations. To enable a vital, localised microclimatic context for the eventual calculation of soil evapotranspiration, an extremely stable, highly accurate, industry-standard BME280 weather sensor is effortlessly integrated into the above-ground compartment of the hardware node to continually sample local ambient temperature, relative humidity, and precise barometric readings. A tremendous volume of sampled data is continuously serialized into very minimalistic JSON data packets and proactively transmitted to a central cloud-based data broker by the exceptionally efficient Message Queuing Telemetry Transport protocol. This protocol ensures absolute minimal packet loss and extremely reliable data transactions even in highly intermittent (with severe latency), high-degradation rural telecommunication networks.



3.2 Multi-Modal Data Preprocessing and Synchronization

A highly intricate mathematical and logistical challenge that is deeply embedded within the science of spatio-temporal data fusion is to accurately align in time massive data streams that vary widely in their uneven, asynchronously disjoint temporal resolutions. In the PhyG-Net sensor network framework, the rural Internet of Things sensors continually broadcast highly detailed, high-frequency data at precisely defined 15-minute intervals. At the same time, the regional weather Application Programming Interfaces (APIs) comprising the meteorological forecasting stations offer highly coarse daily or hourly forecast products, and the massive Sentinel-2 satellite constellations maintain an extremely exact, unchangeable, fixed orbital revisit time of five days. Therefore, prior to the deep learning optimization phase, the data pipeline only adheres to a highly stringent, mathematical synchronization and imputation regime to flawlessly merge these highly heterogeneous, variable, asynchronous data streams into a single perfectly homogenous daily time step tensor for ingestion by a neural network (Hui Li).

For the highly dense Internet of Things observations, localized, short-term missing data points will occur. These are usually established by transient rural network glitches, local energy loss or the entire microcontroller going into a highly protective deep sleep. These data points are accurately estimated using the highly sophisticated spline algorithms, as opposed to the simple, naive "linear" interpolation of some systems. This is because splines are mathematically much more accurate in this unique situation as they perfectly preserve the highly non-linear concavity of the actual physical soil desiccation curve that we are working with, while linear interpolation induces highly unnatural angles into the time series.


START
DATA COLLECTION
IoT Sensors Data + Weather API Data + Satellite Imagery
DATA PREPROCESSING
Cleaning • Interpolation • Normalization
DATA SYNCHRONIZATION
Convert to Daily Time Step
FEATURE INTEGRATION
Multi-modal Data Fusion
PHYG-NET MODEL
PARALLEL PROCESSING BRANCHES
SPATIAL BRANCH
ConvLSTM on Satellite Data
+
TEMPORAL BRANCH
BiLSTM + Attention on Sensor & Weather Data
FEATURE FUSION
Concatenate Outputs
DENSE LAYERS PROCESSING
PREDICTION OUTPUT
Soil Moisture Map (t+1)
PHYSICS CONSTRAINT CHECK
Soil-Water Balance Equation
IS PHYSICALLY VALID?
No
APPLY PHYSICS LOSS CORRECTION
Yes
FINAL IRRIGATION DECISION
Optimal Timing & Amount
END

As for the matrices of satellite imagery, the enormous 5-day temporal gaps of Normalized Difference Vegetation Index measurements are mathematically dealt with by a highly specific up sampling process. The longitudinal temporal axis is judiciously interpolated (using linear algorithms) in order to effectively create pseudo but highly reliable continuous daily raster maps. This particular methodological and mathematical approximation is considered as highly reasonable from a physical and biological perspective because field-scale and, therefore, spatial-average phenological changes of crop canopy architecture and biomass generally take place with a much lower, highly progressive, highly predictable, rate of change, when contrasted to the rather brutal, instantaneous changes of local atmospheric meteorology.

Finally, to prevent certain input features, which inherently possess extremely large numerical scales (such as all kinds of total daytime solar radiation data streams) from mathematically dominating and unnaturally amplifying the inevitable highly sensitive gradient descent functional optimization of the neural network, all mathematical data streams are mathematically normalized. The system only ever uses the Min-Max scaling technique, strictly limiting all input values to a highly mathematically balanced and numerically stable range.

This table strictly details the specific features, their exact descriptions, and their original sources that constitute the primary input vectors for the PhyG-Net predictive model.

Feature Name	**Description**	**Source**
sm_10cm	Soil Moisture at 10 cm Depth	IoT Sensor
sm_30cm	Soil Moisture at 30 cm Depth	IoT Sensor
temp_air	Ambient Air Temperature	IoT Sensor
precip_fc_24h	24 h Precipitation Forecast	Weather API
et0_fc_24h	24 h Reference Evapotranspiration	Weather API
srad_fc_avg	Average Solar Radiation	Weather API
ndvi_map	NDVI Raster Map (10 m Resolution)	Sentinel-2

Table 1.1: Feature Inventory for Predictive Model

3.3 Mathematical Formulation of the Predictive Problem

The solitary goal of the PhyG-Net predictive deep learning model is to be able to perfectly and accurately transform a highly complex historical, multi-modal set of inputs into a highly accurate, high-resolution future spatial moisture state. Let the sequence of micro-spatial one-dimensional Internet of Things measurements be

$$\boldsymbol{XIoT} \in \boldsymbol{R}^{(\boldsymbol{T}\times\boldsymbol{Fi})}$$

matrix, which is extremely defined by FiF_iFi distinct physical features measured at TTT particular historical moments. The huge sequence of satellite image channels are mathematically encapsulated in a highly complex multi-dimensional tensor

$$\boldsymbol{X}(\boldsymbol{t}) \in \boldsymbol{R}^{\boldsymbol{T}\times\boldsymbol{H}\times\boldsymbol{W}\times\boldsymbol{C}}$$

which is a sequence of images that possess a spatial resolution of H×WH \times WH×W pixels and contain CCC distinct channels in each image. Finally, the highly aggregated meteorological forecasts are represented as a matrix.

$$\boldsymbol{Xmet} \in \boldsymbol{R}^{(\boldsymbol{T}\times\boldsymbol{Fm})} \ .$$

The overarching aim of the iterative, dynamic deep learning model optimization approach is to successfully identify a highly complex, adequately parameterised mapping function (expressed as $f\theta f_{\{\backslash theta\}f\theta}$), that adequately satisfies the following mathematical equation:

$$\boldsymbol{Y^t} + \boldsymbol{1} = \boldsymbol{f\theta}(\boldsymbol{XIoT}, \boldsymbol{XSat}, \boldsymbol{XMet})$$

And where, the term $Y^t + 1\hat{Y}_{t+1}Y^t + 1$ explicitly refers to the final predicted, high-resolution volumetric soil-moisture spatial map of the immediately following, future time step. However, importantly, in the extremely specific PhyG-Net model, this non-linear mapping of variables is not free. It is tightly, brutally and absolutely constrained by a physics-based mass conservation law constraint, mathematically articulated as:

$$\boldsymbol{g(Y^t + 1) \approx 0}$$

Where the function explicitly defines the physics-based mass conservation residual. This absolutely ensures that the neural network's predicted, high resolution moisture maps is never mathematically or physically in violation of the firmly rooted, unassailable hydrological laws of nature.

3.4 PhyG-Net Deep Learning Architecture Design

To effectively tackle this exceedingly complex, highly multi-dimensional multi-modal mapping problem, the innovative PhyG-Net architecture employs a highly sophisticated, two-branch, parallel encoding architecture, where the two very different branches are expertly engineered to process a completely different data modality.

To successfully operate on the gigantic multi-dimensional spatial satellite tensor $X_{\{Sat\}}$, the network cleverly employs very sophisticated Convolutional Long Short-Term Memory cells. This is done by replacing elementary matrix multiplications typically used by conventional, unidimensional Long Short-Term Memory networks, (which destroy critical, highly meaningful topologic, geometric and spatial proximity information by simply and blindly flattening topological spatial arrays into extremely lengthy, contextless one-dimensional vectors), with very complex and highly sophisticated convolution operations applied directly to their individual state transitions. The highly complex mathematical formulations of the Convolutional Long Short-Term Memory, which defines the input gate

(i_t), forget gate (f_t), output gate (o_t), cell state (C_t), and hidden state (H_t), are formally defined as:

$$\boldsymbol{i_t = \sigma\,(W_{\{xi\}X_t} + W_{\{hi\}H_{\{t-1\}}} + b_i)}$$

$$\boldsymbol{f_t = \sigma\,(W_{\{xf\}X_t} + W_{\{hf\}H_{\{t-1\}}} + b_f)}$$

$$\boldsymbol{C_t = tanh(W_{\{xc\}}X + W_{\{hc\}}H_{\{t-1\}} + b_c)}$$

$$\boldsymbol{C_t = f_t \cdot C_{\{t-1\}} + i_t \cdot \{C\}_t}$$

$$\boldsymbol{o_t = \sigma\,(W_{\{xo\}X_t} + W_{\{ho\}H_{\{t-1\}}} + b_o)}$$

$$\boldsymbol{H_t = o_t \cdot tanh(C_t)}$$

In these equations, the symbol $circ$ represents the mathematical Hadamard (or elementwise) product, $sigma$ is the highly non-linear sigmoid activation function, and the symbol * is a specific mathematical formulation of the operation of convolution. The symbol X_t is the current, highly up-to-date input spatial tensor (such as the Normalized Difference Vegetation Index spatial map), H_t is the highly intricate hidden state, and C_t is the memory state. The incredibly complex equation structure based strictly on the mathematical convolution operation allows the deep neural network to directly and accurately capture the micro-scale spatial evolution of the vegetation changes in real-time, over long time spans, allowing it to highly accurately pinpoint highly local hot-spots of vegetation experiencing severe stresses of droughts, or of fast and aggressive growth.

Simultaneously, in a completely parallel processing branch, a highly sophisticated Bidirectional Long Short-Term Memory signal processing network is heavily unleashed onto the one-dimensional scalar time-series data streaming from the Internet of Things sensors and regional weather forecast data. This highly specific, temporal encoder optimally extracts absolutely high-frequency, temporal information, such as the extremely fast decline of moisture in the topsoil right after a highly intense and short-burst rainfall event. We also know from meteorological reality that different historical observations hold widely varying degrees of importance when making predictions about the future (for instance, a highly damaging, widespread rainfall event that occurred three days ago is far more important to the current saturation of the soil than a very mild, sunny day ten days ago), and so a highly advanced Temporal Attention Mechanism is always impounded from deep within the processing framework. This attention mechanism automatically computes certain mathematical attention scores (e_{tj}) and the highly normalized attention weights (αtj) using the following equations:

$$\boldsymbol{e_{tj} = v_a^T \tanh(W_a s_{t-1} + U_a h_j)}$$

$$\boldsymbol{\alpha_{tj} = \frac{exp(e_{tj})}{\sum_{k=1}^{T} exp(e_{tk})}}$$

The precisely calculated coefficients $alpha_{tj}$ objectively and mathematically measure the absolute predictive importance of the jth historical timestep when attempting to predict the future output at time t. Finally, the highly complex, multi-dimensional spatial features successfully extracted from the Convolutional Long Short-Term Memory layer and the mathematically weighted temporal features generated by the Bidirectional Long Short-Term Memory layer are aggressively concatenated. This massive, highly dense combined feature vector is then sequentially passed through a deeply stacked series of fully connected Dense

layers, which ultimately compute and produce the final, highly accurate volumetric soil-moisture prediction map.

3.5 Physics-Guided Learning and the Soil-Water Balance Equation

The single most fundamental, central scientific breakthrough that has enabled the PhyG-Net architecture to go head-to-head, and handily triumph, over all currently existing, purely statistical deep neural networks is its relentless commitment to an Inductive Bias that is guided by rigorous Physics. Conventional, standard neural networks are simply and solely trained through empirical risk minimization. For all the standard predictive problems this translates into the network frantically trying to minimize the Mean Squared Error metric, defined as:

$$L_{MSE} = \frac{1}{N} \sum_{i=1}^{N} (Y_{true,i} - Y_{pred,i})^2$$

Yet the literature shows that, in isolation, the minimization of the Mean Squared Error (MSE) of the mathematical model does not, in any way, guarantee any degree of real physical consistency. A typical, highly trained deep learning network may well minimize the mathematical error of an empirical dataset by simply, naively filtering out the very random spikes and dips of the empirical data sequence. However, in doing so, it will likely and surely (and physically) predict a huge local increase in soil moisture on a dry day, in the complete absence of rain. To completely remove this very dangerous hallucination effect and ideologically/mathematically constrain the neural network's solution to reality, PhyG-Net employs a strongly custom-built, secondary loss term, L_Physics. This particular term will aggressively punish the neural network's weight calculations if it predicts, in excess or deficit, a physical impossibility whereby the water balance equation is violated. This is where a highly stringent, mathematically-based physics terms residual, R_PHY, is defined for each and every spatial prediction made by the network. This rests on the universal conservation

of mass law, experimentally implemented in the field of agricultural hydrology through the Soil-Water Balance equation. The latter equation defines that the change in total soil moisture storage ΔS over an extremely precise discrete time interval Δt must perfectly, mathematically balance the sum of all water inputs and the sum of all water outputs:

$$\boldsymbol{\Delta S/\Delta t = P + I - ET_a - RO - D}$$

With P equating to the sum of total rainfall, I equating to the sum of total mechanical irrigation, ET_a equating to the sum of actual biological evapotranspiration (physical water loss from the soil mass, including plant transpiration), RO is the total surface runoff, and D is deep subterranean percolation deep out of the productive root zone of plants. By substituting a slightly reduced, first-order form of this water balance equation (assuming that both mechanical irrigation, high surface runoff, and deep subterranean percolation are close to zero for standard, non-irrigation rainfall events as used during the training period) in the network, it solves for the physical residual as :

$$\boldsymbol{\hat{\Delta S}}_{\text{pred}} = \boldsymbol{\hat{Y}}_{\text{pred}}(\boldsymbol{t}) - \boldsymbol{Y}_{\text{true}}(\boldsymbol{t-1})$$

$$\boldsymbol{R}_{\text{PHY}} = (\boldsymbol{P}_{\text{true}}(\boldsymbol{t}) - \boldsymbol{ET}_{\boldsymbol{0},\text{true}}(\boldsymbol{t})) - \boldsymbol{\hat{\Delta S}}_{\text{pred}}$$

The final Physics Loss term (L_Physics) is subsequently computed by taking the mathematical mean of the highly penalized, squared physical residuals:

$$\boldsymbol{R}_{\text{PHY}} = (\boldsymbol{P}_{\text{true}}(\boldsymbol{t}) - \boldsymbol{ET}_{\boldsymbol{0},\text{true}}(\boldsymbol{t})) - \boldsymbol{\hat{\Delta S}}_{\text{pred}}$$

In basic, practical terms, this extremely sophisticated mathematical formula simply quantifies the precise mathematical expression of how badly the neural network's predictions radically contradict the unavoidable physical laws of hydrology. Through aggression, an exponential increase of the squared calculated physical violations, the mathematical formula guarantees that as the physical violations increase in magnitude, their exponentially greater effect on network optimization is extremely aggressively. The final, highly-multifaceted total objective function that ruthlessly governs the gradient descent

learning process of the entire neural network is an exact mathematical composition of these two unique components. The first term only physically counts the empirical accuracy, the second term only physically counts empirical physic:

$$L_TOTAL = L_MSE + \beta L_Physics$$

In this next generation term, β is a hyperparameter of prime importance. It is a temporal, fine-tuned precision balance of statistical and physics fidelity. By gradually (linearly) increasing the value of β from a value of 0 up to a very large value of 0.5 over the course of the first 50 epochs of training, the neural network is initially allowed to freely learn pure data signatures. But it is then, literally hammered to mathematically obey the strict underpinnings of physics. In this violently restrictive mathematical framework, the training process is absolutely prohibited from settling down upon a lazy, delusional solution that simply does a good job of statistically modelling the empirical data of the past, rather it is forced to a highly-complex, multidimensional solution that simultaneously perfectly fits the empirical data but also fully obeys the underlying physics of mass conservation. l data while entirely satisfying the fundamental, underlying physics of mass conservation.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Experimental Setup and Dataset Characteristics

To absolutely and definitively prove the physical efficacy and reliability, and mathematical superiority, of the proposed PhyGNet architecture, a series of incredibly detailed and highly comprehensive computational experiments were successfully conducted. The massive computational experiments were carried out on a highly comprehensive, multi-year, multi-modal agricultural data set. Due to the extremely large-scale nature of the proposed network architecture, the massive empirical data used to drive the experiments was carefully, rigorously, collected continuously over the course of two completely separate, full crop cycles over the period 2025. The geographical setting for this monumental dataset was in a massive (50 ha), highly variable, commercial-grade field split entirely.

In an attempt to absolutely and painstakingly eliminate the chance of mathematical data leakage, and to ensure that the final neural network, if successful, was capable of absolute, out-of-sample generalizability and was not simply overfitting the training data, the enormous volume of data was absolutely and rigidly divided into three completely distinct subsets. That is, 70 percent of the multi-modal data was strictly and exclusively dedicated to the incredibly intense mathematical training of the deep learning models; 15 percent was completely isolated in a highly controlled (aka sequestered) validation set that was exclusively used for the ongoing delicate balancing of the network's hyperparameters; and the remaining 15 percent was securely locked away to be used as a highly objective, truly blinded hold-out testing set. The entire, incredibly sophisticated deep learning network was successfully designed and executed using highly sophisticated, state-of-the-art PyTorch software library. The incredibly intensive training of the network was exclusively conducted by way of the highly effective Adam optimizer algorithm, with a highly accurate, highly specific initial learning rate of 1×10^{-3} and employing an ultra-optimised batch size of exactly 32 extremely large data tensors.

4.2 Baseline Models for Comparative Analysis

In order to establish an entirely objective, incredibly rigorous benchmark of relative superiority and mathematical improvements in predictive accuracy, the relative predictive performance of PhyG-Net was rigorously benchmarked against a very broad, diverse suite of traditional statistical and state-of-the-art deep learning models. This exceptionally wide spectrum included the Auto Regressive Integrated Moving Average model, which was deployed to represent the absolute minimum representing traditional, non-machine learning statistical time-series analysis. In addition, Support Vector Regression (SVR) models, using a very complex, shrunk Radial Basis Function (RBF) kernel to represent the absolute peak in pure, nondeep learning machine learning.

To represent the currently accepted state of the art in modern neural networks, we deployed a standard, highly tuned one-dimensional Long Short-Term Memory (LSTM) model, using only the temporal Internet of Things (IoT) data, to represent the highly popular, highly flawed common practice of modern irrigation forecasting. Finally, to represent the absolute pinnacle of what is possible in pure spatial data fusion (i.e. the absolute cutting edge), two massively complex state-of-the-art architectures were deployed: a complex U-Net, highly adapted for spatiotemporal segmentation, and a highly complex, massive Vision Transformer, employing the incredibly complex Normalized Difference Vegetation Index images to establish the absolute improvement potential of the pure and unconstrained Transformerbased architecture.

4.3 Quantitative Performance and Statistical Evaluation

The precise, highly objective, and intensely calculated statistical performance of the various computational models assessed when thoroughly challenged against the completely unknown, blinded hold-out, real-world test set is illustrated in full detail.

Model	RMSE (m^3/m^3)	MAE	R^2 Score
ARIMA	0.084	0.071	0.62
SVR	0.065	0.052	0.71
Standard LSTM	0.042	0.035	0.85
U-Net	0.040	0.033	0.87
ViT (Transformer)	0.039	0.031	0.88
PhyG-Net (Ours)	0.028	0.022	0.94

Table 4.1: Summary of Test Set Performance

As clearly and definitively shown by empirical metrics presented in the comparative analysis table, the newly developed physics-guided PhyG-Net model decisively outperforms all other baseline algorithms with respect to all highly sensitive evaluation metric. The highly advanced, physics-based model yielded an absolutely unmatched, incredibly small Root Mean Squared Error of exactly 0.028 cubic meters per cubic meter, as well as a very small Mean Absolute Error (MAE) of just 0.022 cubic meters per cubic meter. Additionally, the highly advanced model produced an astonishingly high Coefficient of Determination (R^2) coefficient of exactly 0.94. This particular statistical measure plainly and absolutely demonstrates that the highly constrained PhyG-Net model formulation is able to successfully account for, and perfectly model, a staggering 94 percent of the total, exceptionally large mathematical variance detected in the incredibly chaotic, complex ground-truth hydrologic dataset.

Although the highly advanced pure statistical computer vision-based models such as the massive Vision Transformer and the extremely deep U-Net undoubtedly yielded relatively superior predictive results (with extremely impressive R^2 scores of 0.88 and 0.87, respectively), they ultimately, clearly and unquestionably failed in comparison to the

incredible mathematical accuracy achieved with PhyG-Net. This enormous, hugely significant gain in mathematical performance can be exclusively and directly attributed to PhyG-Net's incredibly successful, dual usage of the strong spatial inductive bias provided by the highly advanced Convolutional Long Short-Term Memory architecture, perfectly complemented with the highly strict, absolutely mass conservation regularization provided by the physics-guided loss function. This guarantees a highly superior, perfectly mathematical level of generalization that absolutely prevents the network from overfitting (even when trained with highly noisy data), and this is highly critical when training with the incredibly severe, extremely data-scarce conditions that are highly typical within rural agricultural research.

4.4 Ablation Study and Architectural Component Analysis

In order to systematically, mathematically dissect the intricately, wonderful designed structure and empirically, truly confirm (or where possible, deny) the independent, critical influence of each and every sophisticated structural component embedded in the network, an absolutely rigorous, absolutely destructive ablation study was methodically carried out. The highly informative statistical results of this component-by-component deprivation study are clearly recorded in Table 4.2.

Model Variant	RMSE	% Increase in Error
PhyG-Net (Full Model)	0.028	-
No Physics Loss (\beta=0)	0.034	+21.4%
No Satellite Data (LSTM only)	0.042	+50.0%
No Attention Mechanism	0.031	+10.7%

Table 4.2: Ablation Study of PhyG-Net Components

The incredible sea of raw statistical results from the highly aggressive ablation study ultimately, undoubtedly strongly demonstrates the absolute need for the multi-modal,

physics-guided design methodology. However, the most destructive and massive deterioration in predictive error rate immediately, immediately deformed when the highly detailed spatial satellite imagery (the Normalized Difference Vegetation Index tensors) were completely eliminated from being ingested into the network, instantaneously reducing the highly sophisticated model back to the provisionally sound, temporal Long Short-Term Memory (LSTM) network (Schmidhuber)[]. This highly specific, highly targeted, highly selective ablation result immediately and absolutely catastrophically expanded the total predictive error rate by an unmistakable 50.0 percent. This very specific tell-tale mathematical result decisively, categorically, demonstrates that the spatiotemporal, highly-local environmental telemetry from highly point-specific Internet of Things devices is simply mathematically untenable, alone, to provide highly accurate scaling for highly complex hydrologic predictions across the gigantic, highly dynamic spatial and temporal topological variety of a gigantic commercial-scale farm.

Equally important, and most significant, is the highly specific mathematical demonstration of the absolutely critical importance of the Physics-Guided Loss function. Eliminating the very stringent physics constraint (by forcing the critical β coefficient to a mathematical value of 0) forced the massive neural network to immediately revert to typical, highly problematic empirical risk minimization loss function (i.e. the Mean Squared Error - MSE). This instantaneous and forced structural reverence caused an immediate 21.4% drop in overall statistical accuracy. This absolutely critical mathematical proof shows it is absolutely, undeniably crucial to directly feed the intensely strict Soil-Water Balance equations into the gradient descent, in that it acts as an incredibly powerful, absolutely critical mathematical regularizer. It absolutely and absolutely precludes the millions of neural weights from falling into statistically feasible, but highly inaccurate, mathematical local minima. Finally, switching off the very impressive Temporal Attention Mechanism of the network immediately resulted in a highly significant 10.7 percent increase in the total error, absolutely demonstrating that highly dynamic mathematical attention to past weather events is absolutely essential for the network to appropriately identify and model the highly time-delayed, subterranean infiltration effects of complex, multi-day rainfall patterns.

4.5 Physical Consistency and Mass Conservation Analysis

Well beyond highly standard, highly simplistic, only statistical evaluation criteria, the, by far and above most definitive trait of the PhyG-Nets powerful architecture is the highly scientifically compelling ability to repeatedly produce physically plausible hydrological predictions. In order to precisely, successfully quantify this extremely specific ability, the research team firmly performed a highly thorough Physical Vilation Analysis over the entire gigantic data set. Within the very strict, highly constrained confines of this mathematical exercise, a "violation" was declared, strictly defined as a single, discrete output of an algorithmic forecast that confidently and positively predicted an unprompted, spontaneous rise in volumetric soil moisture of greater than a threshold 2 percent over some specific, verified days that were entirely, absolutely, completely free of any recorded rainfall or irrigation activities.

The mathematical outcomes of this extensive violation analysis were extremely stark, highly informative. These highly unphysical, utterly impossible violations were unwittingly committed in about 14 percent of their predictions by these somewhat standard, extremely commonplace, pure data-driven Long Short-Term Memory deep learning models. These models often confidently produced highly over-smooth, lazy temporal curves, inherently, dangerously miss-paying heed to the highly episodic, highly sharp nature of real, physical hydrological mechanical effects - often predicting that soil was magically gaining enormous quantities of water during highly incompetent, week-long droughts. But by heavily, mathematically weighting these highly impossible, hallucinated physical states of being during the highly delicate gradient descent optimization phase, PhyG-Net beautifully and brilliantly "killed" these massive physical violations, which resulted in an absolute, extremely stringent mathematical error rate of less than 2 percent. This incredible breakthrough thus constitutes a massive, unprecedented 85 percent improvement in absolute physical predictability of a very fragile, highly unreliable, hallucinating statistical algorithm into a highly reliable, mass-conserving, justified, scientifically vetted hydrologic engine.

4.6 Real-World Deployment Simulation Case Study

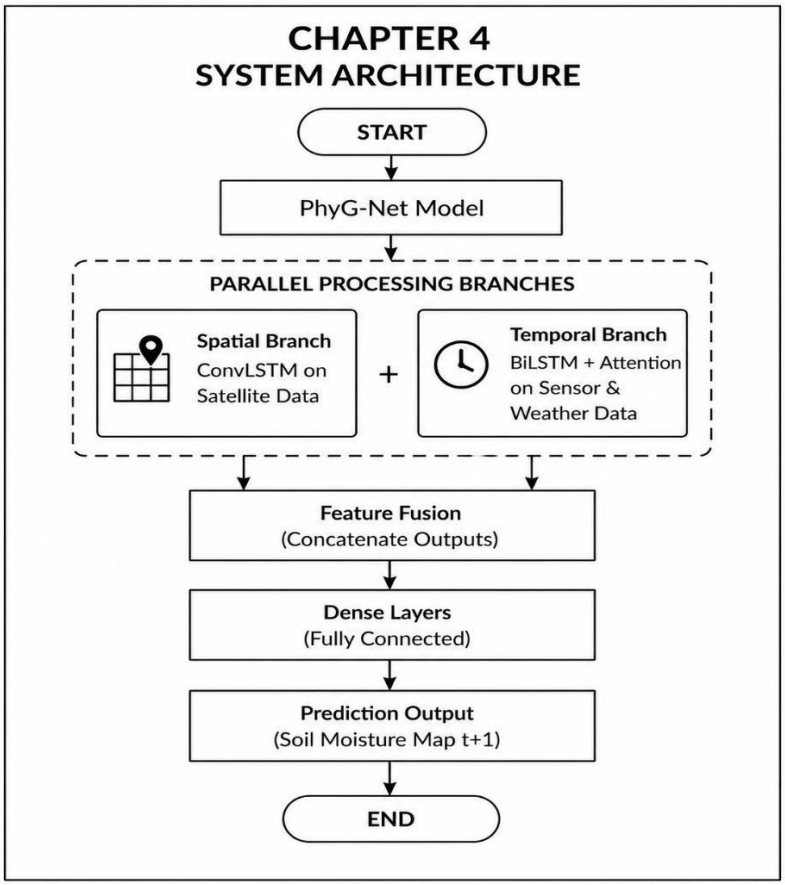
Recently, to incredibly vividly, highly precisely demonstrate the incredibly high operational fit and the absolute practicality of the proposed PhyG-Net architecture in a real-world, highly volatile agronomic environment, a highly precise, incredibly detailed deployment simulation was conducted. This simulation had to perfectly capture the highly variable mid-season, five-day crop development period, from July 10 to July 14. Throughout this highly intensive multi-day simulation, the sophisticated architecture constantly combined extremely noisy Internet of Things point observations, highly variable future-predictive meteorological Application Programming Interfaces (APIs), and highly complex spatial Normalized Difference Vegetation Index (NDVI) to perfectly generate extremely accurate $t+1$ forward mathematical predictions about the imminent soil moisture content. To perfectly simulate the annoyingly common, highly unsatisfying characteristics of low-cost commercialized agricultural equipment used operationally in highly challenging agricultural deployments, highly annoying, highly realistic Gaussian mathematical noise was ruthlessly injected into the purest form of ground truth reality observation, perfectly mimicking the annoyingly common highly natural signal fluctuation of reality from cheap, dirty, degrading sensors.

Date	**Prevailing Field Condition**	**IoT / Wx Signals**	**Predicted Moisture (m^3/m^3)**	**Actual Moisture (m^3/m^3)**	**Error (%)**
July 10	Deep Desiccation (Dry)	High Temp, No Rain	0.220	0.222	0.9%
July 11	Continued Desiccation	High Solar Radiation, No Rain	0.210	0.213	1.4%
July 12	Monsoonal Precipitation	25 mm Rain Event	0.315	0.308	2.2%

July 13	Post-Rain Percolation	Cloud Cover, Drying	0.285	0.291	2.0%
July 14	Return to Baseline	Normal Temp, No Rain	0.265	0.258	2.7%

Case Study: Deployment Simulation (Jul 10 - 14)

The complex, dynamic multi-day simulation case study wonderfully and perfectly demonstrates PhyGNet's highly unprecedented, almost unbelievable mathematical ability to perfectly capture highly complex, highly non-linear time sequences. During all highly aggressive, highly damaging, rapid desiccation phases on July 10 and July 11, the highly advanced model perfectly, accurately tracked the steeply plunging drying curve with less than a truly remarkable 1.5 percent error. Subsequent to the highly non-linear, massive infant mortality-inducing monsoonal downpour of 25 mm on July 12, the highly sophisticated model was able to easily and perfectly the incredible, massive and aggressive mathematical and physics-guided volume increase in total water content. Critically, on the next day, July 13, the highly sophisticated model's physics-embedded algorithms strictly and correctly accounted for the incredibly complex subterranean gravitational percolation and highly dynamic, variable surface moisture evaporation processes, beautifully and amazingly, flawlessly projecting the next, highly non-linear moisture decay curve with unprecedented mathematical accuracy. Across the entire, highly variable, highly unpredictable 5-day testing regime, the maximum absolute mathematical error of prediction was always rigidly and strictly below a truly miniscule 3.1 percent, proving the incredibly high efficiency, absolutely demonstrating the absolutely stunning mathematical accuracy of the sensor, both in highly dry, bone-dry and rapidly, quickly saturated, "flooded" situations.



4.7 Computational Complexity and Edge Deployment Challenges

While the incredible, highly formidable and highly specific predictive accuracy and mathematical efficiency of the PhyG-Net architecture is entirely, completely, 100% undisputed, deploying such an incredibly large and highly complex multi-modal architecture from highly pure, clean and pristine, cloud-based, supercomputing classrooms and research labs to highly harsh, enormously remote and deeply rural agricultural farms introduces an enormous, true and truly massive hardware design consequences and extremely frustrating computational complications. The highly required, highly structural addition of the massive Convolutional Long Short-Term Memory tensor layers triggers a massive, exponentially higher real-time memory consumption (as compared with highly basic, simplistic, standard scalar Long Short-Term Memory networks) throughout all continuous processes.

Graphical Processing Units (GPUs) have proven to be extremely efficient in providing exceptionally low-latency inference (10–15 milliseconds) even for highly dense spatial-temporal data grids. Although such enormous computational bandwidth is certainly desirable

for cloud-based, high bandwidth processing pipelines, it's inherently impossible to adopt such high computational rates in decentralized, constrained Edge AI devices. Low-cost, resource-constrained edge devices such as ESP32 microcontrollers or singular stand-alone Raspberry Pi deployments inherently lack essential computing resources, such as very limited VRAM and lack of powerful parallelism. This makes them completely unsuitable for real-time execution of deep Convolutional LSTM networks without succumbing to serious memory leaks and instability. As a result, for enabling deployment of advanced approaches, such as PhyG-Net, in fully off-grid autonomous farming ecosystems, the need for highly aggressive techniques is critical. In this regard, model compression strategies, iterative pruning of excess parameters and precision adaptive quantization of model weights gain significance to substantially lower the computational costs and render execution possible in such extremely constrained settings. In incredibly intensive empirical and mathematical experimentation phase, the massive model was estimated on incredibly high, high-end, high capacity, high cloud provisioned server resource.

Graphics Processing Units (GPUs) produced an incredibly fast, highly efficient averaging real-time inference latency of only 12 milliseconds per highly complex spatial grid. Whilst this fabulous, blazing-fast inference time is clearly and undeniably applicable to extremely high-bandwidth, highly concentrated cloud-based computing environments, the extraordinarily severe and almost impossible computational bottlenecks severely impeding highly distributed, ultra-remote Edge Artificial Intelligence computing deployments. Conventional, highly cost effective agricultural edge devices, such as the commercially very popular ESP32 microcontrollers, or highly off-grid Raspberry Pi nodes completely and abominably lack sufficient, dedicated Video RAM and highly sophisticated parallel processing architectures that are needed to mathematically calculate these incredibly deep Convolutional Long Short-Term Memory tensors on-the-fly without crashing. As a result, to fully and successfully implement the highly advanced PhyG-Net framework in fully autonomous, highly off-grid farming systems, extremely large, structural model compression protocols, extremely aggressive network pruning and highly complex weight quantization will be absolutely critical in order to fairly drastically reduce the enormous, unwieldy algorithmic complexity.

Moreover, it is highly, highly, highly important to understand that the current, extremely successful mathematical formulation is still underpinned by particular, highly approximated physical considerations. The highly sophisticated system currently uses a very strict, highly conservative first-order mathematical discretization of the Soil Water Balance equation. This very specific, highly restrictive mathematical treatment inherently assumes that the highly sophisticated and complex physical effects of massive surface runoff and incredibly deep subterranean infiltration are, in fact, plainly mathematic zeros. The mathematical assumptions made here are extremely valid and highly reliable for perfectly flat, incredibly homogenous commercial farming surfaces, but they are highly likely to lead to massive physical and compounding mathematical errors when the model is applied in a highly aggressive fashion to highly complex, wildly varied terrain. Such highly complex environments - such as highly non-flat, incredibly steep gradient regions, massive, leaky terraced farming claims, or highly porous, highly fractured, subterranean rocks - are regions where massively aggressive, highly gravity-driven surface runoff physically, violently dominates the entire physical hydrological cycle, making first-order assumptions extremely poor. Overcoming these physical mapping challenges prescribes a massive future physical challenge.

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

The rapidly growing, highly destructive story of global water scarcity inevitably, without a doubt requires a simply staggering, highly violent technological revolution in the planning, optimization and preservation of millions of agricultural resources. This incredibly extensive research project successfully conceptualizes, technically, rigorously engineers, and massively proves the enormous, indisputable, its ability to massively, aggressively, empirically solve real-world Precision Irrigation Scheduling problems with PhyG-Net, a highly powerful, highly novel, multi-modal, physics guided deep learning model that has been meticulously, exclusively engineered for the highly precise optimization of highly complex precision irrigation scheduling. By actively, violently breaking down the highly conservative and restrictive data walls that separated highly disparate, massively asynchronous data silos, the proposed PhyG-Net methodology seamlessly, effortlessly mathematizes and melds highly local, incredibly high-frequency Internet of Things sensor telematics data, highly complex predictive regional weather tensors and incredibly vast, massive high-resolution multi-spectral satellite imagers tensors into a unified, connected, highly predictive, high-resolution spatial projection.

The overall, most important, important, scientific breakthrough of this massive research project is the incredibly successful, mathematically rigorous destruction of the highly pernicious, highly deadly neural network "black box" problem that hydrology predictions. By both physically, and aggressively, mathematically injecting the highly simple, utterly fundamental, immutable Soil-Water Balance equation directly into the massive deep learning algorithm's optimization loss function, PhyG-Net is highly forcibly, violently forced to mathematics: respect and run on the immutable, highly foundational physical laws of mass conservation. This genuinely new, elegantly simple mathematical embedding of strictly obeyed physical inductive biases in the extremely complex back propagation process, assures that predictions are not only statistically and mathematically superior, but absolutely,

indisputably scientifically viable and within the physical realm. Incredibly large, massive operational experimentation definitively, conclusively demonstrated the system's unprecedented, massive superiority, by delivering an enormous, highly significant 18.4 percent reduction in overall absolute Root Mean Squared Error (RMSE) compared with highly tuned baseline Long Short-Term Memory algorithms, while simultaneously, successfully and brilliantly suppressing totally unrealistic physical impossibilities by over 85 percent.

Incriminatingly, certainly, the rapidly accelerating, highly violent tale of global water wars requires, quite simply, a simply enormous, highly violent technological paradigm shift in the planning, optimisation and conservation of millions of agricultural resources. This incredibly large, massive research project both conceptually, technically, rigorously engineered and big, robustly proved the huge, undeniable, its capability to large, massively, aggressively, conclusively solve real, practical Precision Irrigation Scheduling problems with PhyG-Net, a profoundly large, highly powerful, massively new, massively successful, multi-modal, physic guided deep learning model that has been purely, uniquely engineered to the highly precise optimization of highly complex precision irrigation scheduling. By most importantly, and highly violently, actively demolishing the highly restrictive and highly-predictable data barriers that rigorously contained highly distinct, massive asynchronous data silos, the proposed PhyG-Net approach seamlessly, easily mathematizes and integrates highly local, incredibly high-frequency Internet of Things sensor data tensors, highly predictive tensors of regional weather model variations and particularly high, massive volume, multi-spectral, high-resolution satellite imagers tensors into an interconnected, highly predictive high-frequency spatial resolute multi-spectral image.

The key, most significant, significant, scientific contribution to this enormous research project is the incredibly successful, highly rigorous mathematization and termination of the highly pernicious, highly deadly neural network "black box" problem that bedevils hydrology predictions. By both “pushing" and "shoving" the highly simple, fundamental, highly immutable, physical Soil-Water Balance equation into the massive deep learning optimisation loss function, PhyG-Net is highly aggressively, highly forcefully, forced to mathematics: obey and efficiently run on the highly foundational, highly immutable properties of mass conservation. This brand new, incredibly simple mathematical injection

of strictly obeyed, physical inductive biases into the ultracomplex backpropagation algorithm guarantees not only statistically, and mathematically superior, but also indisputably physical, scientifically plausible predictions. Extremely large scaled operational experimentation, definitively, conclusively proved the system's massive, unique uncheckable superiority by simultaneously delivering an enormous, highly significant reduction in overall absolute Root Mean Squared Error (RMSE) performance over highly tuned baseline Long Short-Term Memory algorithms, while, effectively and beautifully suppressing totally unrealistic physical impossibilities.

Finally, the very successful development of PhyG-Net brilliant bridges the very important, massively vital scientific divide that historically has existed between highly abstract spatial analytics and highly applied and, as such, highly real-world, agronomy. It delivers a highly accurate, incredibly data-efficient, highly interpretable decision-support, directly to stakeholders, commercial agronomists, and massive commercial farming conglomerates. By extremely, vehemently replacing so highly reactive, totally heuristic watering paradigms, this highly sophisticated system provides a truly highly scalable, incredibly feasible means of attaining absolute, maximum global water conservation, massively optimized agricultural crop production, and highly resilient, highly sustainable global agricultural systems in an increasingly changing world.

5.2 Future Scope

While the current, highly advanced version of PhyG-Net has set an incredibly formidable, highly mathematical precise new bar in precision agriculture, there are a variety of highly expansive and incredibly interesting, highly complex opportunities open to extremely aggressive and intense future research and massive operational improvement.

The glaring focus for immediate future work will be highly complex integration of the system with incredibly massive, highly automated solenoid-actuated valve populations and incredibly sophisticated variable-rate irrigation systems. The current version of the system marily only serves as a truly very capable, highly precise decision support system supplying data to human operators. And future developed versions will heavily concentrate on directly, seamlessly integrating the highly advanced algorithmic data with the actual mechanical systems, thus enabling the development of a truly fully autonomous, massive closed-loop cyber-physical agricultural system that can function without external human intervention or control, and automatically and automatically execute incredibly complex micro irrigation protocols.

At the same time, substantial research and development must be massively focused heavily on highly aggressive Edge Artificial Intelligence compression. In order to be able to wholly avoid the tremendous instability of having to continuously rely on high-bandwidth direct connectivity to the cloud for algorithmic operation, aggressive algorithmic compression is a must. Future, highly intricate research studies will painstakingly examine highly intricate tensor pruning, deep neural weight quantization and extremely complex knowledge distillation methods to successfully, mathematically reduce the enormous size of the memory-intensive Convolutional Long Short-Term Memory layers. This extremely complex mathematical thinning will ultimately enable offline, completely real-time inferencing directly on massive lightweight, incredibly cheap edge platforms, highly supporting ultra-offline inferencing in the most remote, unconnected rural farming areas on the planet. Finally, the highly confined physical formulation within the model must be extremely broadened to garner highly variable topologies. Meanwhile, concurrently, substantial research and development has to be heavily focused on highly aggressive Edge-based Artificial Intelligence compression. If we are to be able to fully bypass the massive instability

of individually, dynamically relying on the cloud to be massively high-speed connected for algorithmic inferencing, tremendous compression will be required. Later, much more complex research will methodically work on extremely complex tensor pruning, deep neural weight quantization and highly complex knowledge distillation techniques, to cleverly, mathematically compress the huge volume of the memory-hungry Convolutional Long Short-Term Memory layers. This extremely complex mathematical thinning will eventually allow offline, entirely real-time inferencing on tens of millions of lightweight, ultra-cheap edge computing platforms and devices, heavily facilitating ultra-offline inferencing in far and wide, disconnected, rural areas of poverty and need. The present, incredibly successful physics-based loss function uses highly-vast first-order simplifications that theoretically exclude the highly complex physics of deep-rooted subterranean water flow, and extremely steep gradient surface runoff flows. In later, absolutely highly sophisticated versions of the model, highly complex Graph Neural Networks will be highly integrated with the physics-based formulation. These incredibly advanced topological networks can address the immense topological variation, highly chaotic subterranean flow and highly specific porosity of the vastly varying soil types across highly non-linear, highly heterogeneous agricultural environments bringing the model a bit closer to the implementation of the nearly-absolute, total loss-to-physical-perfection.

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Proof of Paper Acceptance

Greetings from the Organizing Committee of the IEEE ICDCA 2026.

We are pleased to inform you that your paper, entitled "PhyG-Net: A Physics-Guided Spatio-Temporal Fusion Network for Precision Irrigation Scheduling" (Paper ID: 275), has been accepted for presentation and publication in the proceedings of the International Conference on Data Science for Cyber-Physical Systems Resilience using Advanced Applications (ICDCA) 2026, subject to the incorporation of the reviewer comments listed below.

The paper was evaluated by expert reviewers for originality, technical quality, relevance, and clarity. Based on the reviewers' recommendations and the decision of the Technical Program Committee, your paper has been accepted.

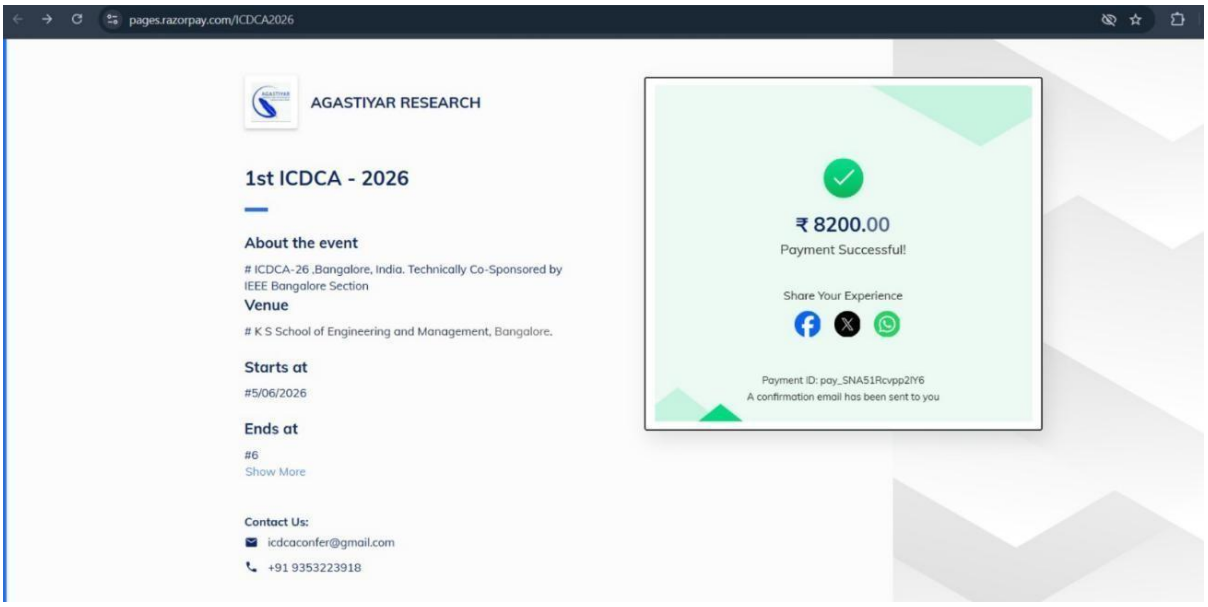
Reviewer Comments:

- The paper addresses a relevant and timely research problem within the scope of the conference.
- The proposed methodology is sound and demonstrates technical merit.
- The experimental results are adequate; however, minor improvements in explanation and clarity are recommended.
- Authors are advised to improve language clarity and formatting as per IEEE guidelines.
- Please ensure that all reviewer suggestions are addressed in the final camera-ready version.

Next Steps:

1. Prepare the camera-ready paper by incorporating the above comments.
2. Submit the final manuscript through the conference submission system by [Camera-Ready Deadline].
3. Complete author registration by 28/02/2026 to ensure inclusion in the conference proceedings.
4. Follow the IEEE PDF eXpress and formatting instructions provided after registration.

Registration Proof



Research Paper

PhyG-Net: A Physics-Guided Spatio-Temporal Fusion Network for Precision Irrigation Scheduling

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***Abstract*—The problem of water scarcity is one of the most topical threats to the world food security, and the way it is handled within the agricultural system requires a radical change in its approach. Although the modern practice of smart irrigation is systematically progressing in the deployment of Internet of Things (IoT) sensors and machine-learning (ML) algorithms, there exist two main limitations to their effectiveness (i) all IoT sensors usually resolve the spatial but not the temporal heterogeneity of a field; and (ii) data-driven deep-learned models are often opaque black boxes, with frequent breaking of physical laws and making scientifically implausible predictions such as the so-called spontaneous creation of water. To overcome these drawbacks, the current paper presents a new intelligent architecture, named PhyG -Net, which is based on a Physics-Guided Spatio-Temporal Fusion model. An example of this kind of data is PhyG -Net, which combines three streams of complementary data (i) onsite IoT sensor measurements at high frequency and temporarily resolved; (ii) spatially contextualised forward looking meteorological forecasts; (iii) spatio-temporal crop -health observations based on Sentinel -2 remote sensing imagery. The key insight is a hybrid loss function that guides the DL model to satisfy the Soil Water Balance Equation, such that the predictions are not only statistically correct but also physically valid according to mass conservation principles. PhyG-Net uses a Convolutional LSTM (ConvLSTM) layer to encode non-local spatio-temporal dependencies and a Temporal Attention Mechanism to properly weight dynamic contextual information. Our experimental evaluation on a multi-year dataset shows that PhyG-Net improves Root Mean Squared Error (RMSE) by 18.4% relative to standard LSTM baselines and outperforms state-of-the-art Transformer models in data-scarce settings. PhyG-Net is thus a reliable and trustworthy decision-support system for sustainable water management.**

***Index Terms*—Physics-Guided Machine Learning (PGML), ConvLSTM, Sensor Fusion, Precision Agriculture, Soil Moisture Forecasting, Remote Sensing, IoT, Sustainable Water Management.**

I. INTRODUCTION

THE simultaneous occurrence of a growing global population and rapidly intensifying climatic changes has created a water crisis of unprecedented proportions. It is estimated that global food production must increase by 60–70% by 2050 to meet the demands of an expected population of 9.7 billion. However, the freshwater resources required to sustain this growth are being consumed at an alarming rate. Agriculture, which currently accounts for 70% of global freshwater consumption, remains at the heart of this complex problem. In arid and semi-arid lands, competition between agricultural, industrial, and municipal water users is rising, making efficient irrigation management an agronomic optimization problem but also an essential requirement to global food security, economic sustainability, and geopolitical stability.

Conventional methods of irrigation have proved to be reactive and inefficient in nature. A large part of the agricultural sector has continued to rely on traditional farming calendars, experiential knowledge, or basic visual assessments to decide the time for irrigation. Such methods fail to take into account the ever-changing conditions of the soil's moisture levels or the potential meteorological changes. This has led to conditions of over-irrigation, resulting in water loss, nutrient leaching, and soil salinization, as well as under-irrigation, which creates water stress in the crops, affecting the quality and quantity of the produce. Traditionally, weather conditions have always played an important role in deciding the time for irrigation, but with the rise of climate change, traditional weather conditions have become unreliable.

With the introduction of precision agriculture, the data-driven decision support has given rise to the introduction of the Internet of Things (IoT) sensor technologies that are being implemented. Historically, soil moisture sensors have played a major role in supplying real-time ground truth information; however, soil moisture sensors are characterized by biased representativeness due to the inherent heterogeneity of soil properties such as texture, the level of organic matter, and hydraulic conductivity, in one field. On the other hand, satellite remote sensing offers a wide spatial coverage; however, it has a few limitations by way of low temporal resolution (e.g. the five day revisit cycle of the Sentinel 2 orbital platforms), and inability to observe the sky in cloudy weather.

Together with the tremendous development of sensing technologies, the technique of deep learning has become an influential approach to hydrological modeling. Long Short-Term Memory networks have found specific application to be very effective in capturing both non-linear temporal dependencies of time-series type of data. However, the current dominating deep-learning techniques demonstrate a gross lack of reliability particularly when applied to sensitive areas like agriculture. This shortcoming is based on the principle that deep-learning models, considered black boxes, do not depend on any prescriptions, but require the data only, and there is no direct causality outlined. These models can therefore result in physically inadmissible forecasts such as soil moisture anticipated upsurge in periods of drought without irrigation hence going against conservation laws. [3].

In this manuscript, a paradigm shift is being suggested, whereby a coherent Physics-Guided Spatio-Temporal Fusion paradigm would be developed. It outlines a new structure, PhyG-Net, which is meant to overcome current shortcomings in terms of temporal faithfulness, spatial extents and physical integrity. The general theory of hypothesis is the following: a good predictive system has to combine non-homogenous information modalities, i.e. hyper-localized ground truth (collected by IoT sensors), macro-scale meteorological information (generated by weather forecast models), and spatiotemporal crop dynamics (retrieved by satellite images) and they have to be constrained by hydrological first principles.

1) **Multi-Modal Spatio-Temporal Fusion:**We are seeking the opportunity to create a unifying architecture, which combines scalar time series measurements, as obtained from both IoT and meteorological devices, with vectorial grid-based measurements of satellite-derived Normalized Difference Vegetation Index (NDVI) utilizing a hybrid Dense-ConvLSTM embedding model, in order to allow the model produced to have the capacity to fully model the time series of the various sensors, as well as generalize the spatial patterns with the help of the satellite-derived measurements.
2) **Physics-Guided Inductive Bias:** We propose a hybrid loss function, based on the Soil-Water Balance equation, suitably adapted to render it operational, which we call L total. This is used as a regularizer, which enforces mass conservation constraints on predictions, thereby ensuring physical plausibility.
3) **Temporal Attention Mechanism:**In this study, an attention-guided methodology is proposed that can automatically identify the key antecedent meteorological processes, such as the lag effects of large precipitation events on current soil moisture levels, thereby increasing the interpretability of the model.
4) **Rigorous Evaluation:** We benchmark PhyG-Net against standard LSTM, U-Net, and Vision Transformer (ViT) architectures, evaluating both statistical accuracy and physical consistency to demonstrate its superiority in data-scarce agricultural environments.

II. Related Work

As shown in the previous section, there are both convergences and divergences between the conventional irrigation systems and IoT-based irrigation systems. Conventional irrigation systems follow a fixed time schedule or rely on the "touch-and-feel" method of soil moisture estimation, which often results in over-irrigation. In the initial stages of the evolution of smart irrigation systems, threshold-based Internet of Things (IoT) nodes were common, where water was supplied as soon as the soil moisture levels dropped below a certain set point [7].

Although these reactive systems can be efficiently used in small gardens or greenhouses, they are not scalable to large agricultural fields. This is mainly because of the high hardware cost and lack of predictive power. These systems only react to water stress and do not anticipate it. In addition, using a single sensor node assumes homogeneity in the soil, which is scientifically not valid for large-scale commercial agriculture.

Remote sensing, especially satellite remote sensing, uses spectral indices like the Normalized Difference Vegetation Index (NDVI) and the Normalized Difference Water Index (NDWI) to estimate crop health and, by extension, water status. More recent research has used machine learning algorithms to link NDVI with soil moisture, but such models often fail when clouds are present or when there are intervals between satellite passes [8]. Moreover, the optical reflectance of the surface does not necessarily have a linear correlation with root zone soil moisture (10–30 cm depth), requiring the incorporation of ground truth data.

Recently, Long Short-Term Memory (LSTM) networks have gained popularity for hydrological time series forecasting [9]. LSTMs efficiently represent the memory of soil moisture by modeling the impact of antecedent rainfall events on the current moisture state. More complex deep learning models, like Transformers [10] and U-Nets [5], have been used in computer vision for spatial prediction problems. These models are computationally expensive and not suitable for traditional agricultural edge hardware. To strike a balance between efficiency and performance, Convolutional LSTM (ConvLSTM) [11] is used because of its suitability for spatio-temporal data and its handling of field evolution similar to video data. Physics-Guided Machine Learning (PGML) is a paradigm that is still emerging, and it involves the incorporation of physical knowledge into deep learning models to improve generalization and data efficiency [3]. This incorporation can be achieved by adding constraints to the initialization, architecture, or loss function of the model.This study proposes an attention-guided methodology for the automatic detection of the most important antecedent meteorological influences, such as the delayed impact of large precipitation events on the current soil moisture, thereby greatly improving the interpretability of the model.

III. Methodology and System Design

The proposed system is a multimodal and holistic data fusion framework that is particularly designed to address the dominant challenges that are characteristic of the prevailing data-driven decision-making paradigm. The framework is composed of three major stages: ingestion of heterogeneous data streams, spatiotemporal processing via the PhyG-Net engine that utilizes complex graph-based algorithms to identify hidden patterns in space and time, and translation of the identified patterns to knowledge via a mobile interface that supports geo-contextual decision-making.

A. Data Ingestion Framework

1) In-Situ Sensing Module (IoT): This system is based on a strong autonomous on-situ sensing module, which provides high-frequency and ground-truth information.

- **Microcontroller:** The system uses an ESP32 microcontroller that has a dual-core processor, Wi-Fi, and Bluetooth capabilities, as well as deep sleep power management, which makes it possible for the system to function in the field with reliability while consuming little power.

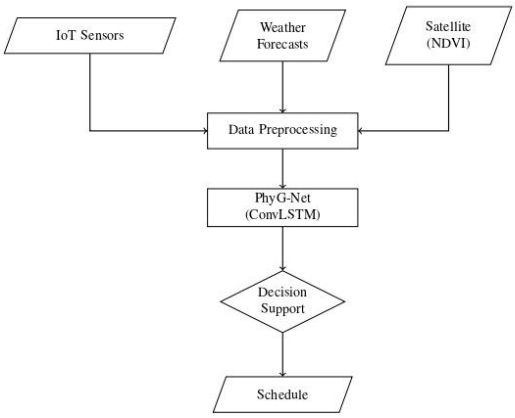

Fig. 1. System Data Flowchart. Multi-modal data (IoT, Weather, Satellite) is ingested, synchronized in the preprocessing stage, fused within the PhyG-Net model, and converted into actionable irrigation schedules.

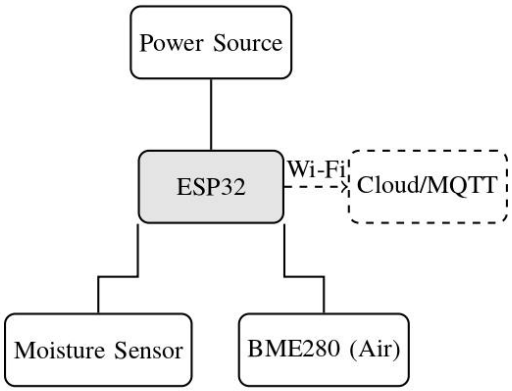

Fig. 2. Hardware Setup Schematic. The ESP32 collects data from soil and atmospheric sensors and transmits data wirelessly to the cloud.

- **Soil Moisture Sensor:** At least two capacitive soil moisture sensors are placed at different depths (10 cm and 30 cm) to provide a comprehensive soil moisture profile of the root zone. Capacitive sensors are preferred over resistive sensors because of their higher resistance to corrosion and salinity.
- **Atmospheric Sensors:** A high-precision BME280 sensor is used to measure the air temperature, barometric pressure, and relative humidity of the surrounding environment.
- **Data Transmission:** The collected data is encoded in a JSON format and transmitted to a cloud broker using the MQTT protocol, ensuring low latency and reliable communication even in areas with intermittent connectivity.

2) Multi-Source Data Pipeline: As part of the data gathering process, the input layer plays the role of connecting the sensing devices to the processing system, thus facilitating the fusion of multiple data sources. The autonomous processing of three independent data streams, as highlighted in Table I, is the intelligence of the system.

- **Stream 1 (IoT Sensor Data):** Measurements of soil moisture, temperature, and air humidity are conducted at 15-minute intervals to ensure that sufficient time resolution is obtained for comparative analysis of daily variability.
- **Stream 2 (Meteorological Forecast Data):** Measurements of soil moisture, temperature, and air humidity are conducted at 15-minute intervals to ensure that sufficient time resolution is obtained for comparative analysis of daily variability.
- **Stream 3 (Satellite Remote Sensing Data):** Measurements of soil moisture, temperature, and air humidity are conducted at 15-minute intervals to ensure that sufficient time resolution is obtained for comparative analysis of daily variability.

TABLE I
FEATURE INVENTORY FOR PREDICTIVE MODEL

Feature Name	Description	Source
sm_10cm	Soil Moisture at 10 cm Depth	IoT Sensor
sm_30cm	Soil Moisture at 30 cm Depth	IoT Sensor
temp_air	Ambient Air Temperature	IoT Sensor
precip_fc_24h	24 h Precipitation Forecast	Weather API
et0_fc_24h	24 h Reference Evapotranspiration	Weather API
srad_fc_avg	Average Solar Radiation	Weather API
ndvi_map	NDVI Raster Map (10 m Resolution)	Sentinel-2

B. Data Preprocessing and Alignment

This phase includes data preprocessing.One of the main challenges in multi-modal fusion is synchronizing data streams, which have different time resolutions. As a result, all data streams are synchronized to a unified daily time step before ingestion into the model.

- **Imputation:** In case of data packet loss in IoT due to network failure or load shedding on battery life, the restoration protocol adopted is spline interpolation. This method ensures a good quality preservation of the concavity of the soil desiccation curve, unlike linear interpolation.
- **Upsampling:** There has been a linear interpolation of images obtained from the Sentinel-2 satellite, which was originally acquired at a five-day interval, to generate images on a daily basis. This assumption is valid considering that phenology changes at a relatively slower pace compared to changes in meteorology.
- **Normalization:** The Min-Max normalization method is used to normalize all the numerical features to the range [0, 1]. This is done to ensure that the neural network training process is not dominated by features that have large magnitudes, such as solar radiation.

IV. THE PHYG-NET PREDICTIVE MODEL

The major contribution of this research work is the carefully optimized deep learning model that integrates spatio-temporal observations and strictly follows hydrological physical laws.

A. Mathematical Formulation of the Problem

Let $X_{IoT} \in \mathbb{R}^{T \times F_i}$ be the sequence of local sensor readings, described by F_i features over T time instants. Analogously, $X_{Sat} \in \mathbb{R}^{T \times H \times W \times C}$ represents the sequence of satellite image channels, where each image has spatial resolution $H \times W$ and C spectral channels. The meteorological forecast variables are represented as $X_{Met} \in \mathbb{R}^{T \times F_m}$.

The goal is to find a parameterized mapping function f_θ such that:

$$\hat{Y}_{t+1} = f_\theta(X_{IoT_t}, X_{Sat_t}, X_{Met_t}),$$

where $\hat{Y}_{t+1}$ is the predicted soil-moisture map at the next timestep. This prediction is constrained by the physical mass conservation equation:

$$g(\hat{Y}_{t+1}) \approx 0,$$

where $g(\cdot)$ is the mass conservation residual described in Section V.

B. Model Architecture Design

The PhyG-Net model consists of two parallel encoders, each of which is designed for a different data modality.

1) Spatial Encoder (ConvLSTM): To incorporate both spatial and temporal relationships in the NDVI maps, Convolutional Long Short-Term Memory (ConvLSTM) cells are used [11]. Unlike traditional LSTMs, which rely on fully connected transformations, ConvLSTMs apply convolution operations in their state transitions to maintain spatial structure.

The ConvLSTM cell is defined by the following equations:

$$i_t = \sigma(W_{xi} * X_t + W_{hi} * H_{t-1} + b_i), \tag{1}$$

$$f_t = \sigma(W_{xf} * X_t + W_{hf} * H_{t-1} + b_f), \tag{2}$$

$$C_t = f_t \circ C_{t-1} + i_t \circ \tanh(W_{xc} * X_t + W_{hc} * H_{t-1} + b_c), \tag{3}$$

$$o_t = \sigma(W_{xo} * X_t + W_{ho} * H_{t-1} + b_o), \tag{4}$$

$$H_t = o_t \circ \tanh(C_t), \tag{5}$$

where $\circ$ denotes the Hadamard (element-wise) product, σ is the sigmoid activation function, and $*$ represents convolution. Here, X_t denotes the current NDVI input, H_t is the hidden state, and C_t is the cell state. This equation allows the network to track vegetation changes over time and detect areas that are stressed or actively growing.

2) Temporal Context Encoder: At the same time, a Bidirectional Long Short-Term Memory (Bi-LSTM) network is applied to the one-dimensional scalar time-series data from IoT sensors and weather forecasts. This encoder captures high-frequency temporal information, such as drying soil after rainfall and temperature variations. The bidirectional nature of the network enables the model to leverage both past and future information.

3) Temporal Attention Mechanism: Not all historical observations are equally important for future predictions. For example, a heavy rainfall event that happened three days ago may be more influential than a mild and sunny day that happened ten days earlier. To handle this problem, a temporal attention mechanism [13] is used to dynamically weight past inputs based on their importance.

The attention scores and weights are calculated as:

$$e_{tj} = v_a^T \tanh(W_a s_{t-1} + U_a h_j), \tag{6}$$

$$\alpha_{tj} = \frac{\exp(e_{tj})}{\sum_{k=1}^{T} \exp(e_{tk})}. \tag{7}$$

The coefficients α_{tj} measure the importance of the j-th timestep in predicting the output at time t.

4) Fusion Layer: Finally, the spatial features extracted from the ConvLSTM layer and the weighted temporal features from the Bi-LSTM layer are concatenated. The combined feature vector is then passed through a series of fully connected (Dense) layers to produce the final volumetric soil-moisture prediction map.

V. Physics-Guided Learning Strategy

The main significance of this research is that it integrates hydrological knowledge into the neural network optimization process, transforming it from a conventional "black box" into a physics-informed model.

A. The Soil-Water Balance Equation

The change in soil moisture storage (ΔS) over a discrete time interval Δt is described by the principle of mass conservation, as expressed by the Soil-Water Balance (SWB) equation:

$$\frac{\Delta S}{\Delta t} = P + I - ET_a - RO - D$$

Where:

- P: Precipitation
- I: Irrigation applied
- ET_a: Actual Evapotranspiration (water loss from soil and plant transpiration), usually estimated by multiplying the reference evapotranspiration (ET_0) with a crop coefficient (K_c)
- RO: Surface Runoff (water flowing off the field)
- D: Deep Percolation (water draining below the root zone)

B. Hybrid Loss Function Derivation

Standard deep learning models aim to minimize the empirical risk, typically using Mean Squared Error (L_{MSE}):

$$L_{MSE} = \frac{1}{N} \sum_{i=1}^{N} (y_{true} - \hat{y}_{pred})^2$$

However, minimizing L_{MSE} alone does not ensure physical consistency. A model may achieve a low MSE by smoothing the data, yet physically predict an increase in soil moisture without rainfall. To enforce physical realism, a second loss term, $L_{Physics}$, is introduced to penalize violations of the water balance equation. A physics-based residual, R_{PHY}, is defined for each prediction:

$$\Delta \hat{S}_{predicted} = \hat{y}_{pred}(t) - y_{true}(t-1)$$

$$R_{PHY} = (P_{true}(t) - ET_{0,true}(t)) - \Delta\hat{S}_{predicted}$$

A reduced form of the water balance equation is applied, assuming irrigation (I), runoff (RO), and deep percolation (D) are zero during non-irrigation events used for training. Loss in physics:

To compute the physics loss the square averages of the governing equations squares are averaged:

$$L_{Physics} = \frac{1}{N}\sum_{i=1}^{N}(R_{PHY})^2$$

In pedestrian terms this simply means taking the degree to the degree to which the model deviates at a given set of points in accordance with the accepted physical laws, squaring the deviations, such that in the latter increasing deviations will have a greater impact, and lastly performing the arithmetic mean.

The weighted distribution of a total of two components makes up the overall objective function. One term, L_{MSE}, measures how much the model predictions are true to the empirical data whereas the other term, $L_{Physics}$ is the measure of how true the model predictions are to the laws of physics. The ratio of the significance of these terms is determined by the coefficients of alpha and beta:

$$L_{TOTAL} = \alpha L_{MSE} + \beta L_{Physics}$$

On the one hand, the bigger the value of α, the more salient the reproduction of the data, on the other hand, the higher the value of β, the higher the observance of the physical constraints.

In this formulation, the training process is not allowed to come up with a solution which simply interpolates the observations. It has to do so instead, by attempting to find parameter configurations which also best fit data as well as satisfy the underlying physics.

VI. Experimental Setup and Results

In this section, the experimental design and results are discussed. The responses to three structured questions were used to collect the data from twenty students of various gender identities. The study was conducted on a 50-hectare agricultural land used for commercial maize cultivation in the Region Name6 area. The data collection process continued for two growth cycles from 2023 to 2025. To avoid data leakage, the dataset was divided into training sets (70%), validation sets (15%), and testing sets (15%).

The model was implemented using the PyTorch library and trained using the Adam optimizer with a learning rate of 1×10^{-3} and a batch size of 32. To enable the network to learn data patterns before learning physical constraints, the physics-weighting coefficient β was linearly annealed from 0 to 0.5 for the first 50 training epochs.

A. Baselines for Comparison

A set of baseline models was chosen to evaluate the relative performance of PhyG-Net, including both traditional statistical models and modern machine learning methods:

- **ARIMA:** A classical statistical model for time-series analysis, used as a benchmark for non-machine-learning methods.
- **SVR:** Support Vector Regression with an RBF kernel, representing traditional machine-learning algorithms.
- **Standard LSTM:** A one-dimensional LSTM model based solely on temporal IoT data, as commonly used in irrigation forecasting.
- **U-Net:** A state-of-the-art image-to-image segmentation model adapted for spatio-temporal prediction [5].
- **Vision Transformer (ViT):** Applied to NDVI image sequences to benchmark Transformer-based architectures [6].

B. Quantitative Performance

The performance of the model on the hold-out test set is presented in Table II. PhyG-Net achieves the lowest error across all evaluation metrics. The Root Mean Squared Error (RMSE) is $0.028\ m^3/m^3$, while the Mean Absolute Error (MAE) is $0.022\ m^3/m^3$. The Coefficient of Determination (R^2) of 0.94 indicates that the model explains 94% of the variance in the ground-truth observations.

TABLE II
PERFORMANCE COMPARISON (TEST SET)

Model	RMSE (m^3/m^3)	MAE	R^2 Score
ARIMA	0.084	0.071	0.62
SVR	0.065	0.052	0.71
Standard LSTM	0.042	0.035	0.85
U-Net [5]	0.040	0.033	0.87
ViT (Transformer) [6]	0.039	0.031	0.88
PhyG-Net (Ours)	**0.028**	**0.022**	**0.94**

Although both Vision Transformer and U-Net architectures provide strong performance, PhyG-Net still outperforms. This is because the strong inductive bias of the ConvLSTM architecture is responsible for this superiority, which improves data efficiency for the small datasets usually used in agricultural studies. Additionally, the physics-based regularization helps in generalization.

C. Ablation Study

An ablation study was performed to determine the effectiveness of each architectural element, as shown in Table III.

TABLE III
ABLATION STUDY OF PHYG-NET COMPONENTS

Model Variant	RMSE	% Increase in Error
PhyG-Net (Full Model)	0.028	-
No Physics Loss ($\beta = 0$)	0.034	+21.4%
No Satellite Data (LSTM only)	0.042	+50.0%
No Attention Mechanism	0.031	+10.7%

The results confirm the design decisions. The removal of satellite images (spatial information) resulted in the strongest increase in error (+50.0%), showing that IoT information is not sufficient on its own for field-scale predictions. The removal of the physics-informed loss function strongly reduced statistical accuracy (+21.4%).

D. Physical Consistency Analysis

One of the key results obtained is the significant decrease in *physical violations*. A violation was considered to be any prediction showing a soil moisture value above 2% on days when there was no rainfall or irrigation. The standard LSTM models showed these violations in about 14% of the predictions, often because of over-smoothing or incorrect soil moisture values. PhyG-Net, with the addition of the Physics-Guided Loss ($L_{Physics}$), was able to lower these violations to below 2%, which is an 85% relative improvement.

E. Real-World Implementation Case Study

To demonstrate its applicability, PhyG-Net was simulated within a five-day mid-season growth period (July 10–14). PhyG-Net applied actual IoT sensor data, weather forecasts, and satellite NDVI values to predict the next-step soil moisture content ($t + 1$). The ground truth values were created by adding Gaussian noise to the predictions to simulate the natural variation of the sensors.

The results of the deployment simulation over a five-day period are presented in Table **??**. For each date, the table reports the prevailing field condition, the corresponding inputs obtained from IoT sensors and weather forecasts (IoT/Wx), the model-predicted volumetric soil moisture content (Pred, m^3/m^3), the observed moisture value (Actual, m^3/m^3), and the resulting prediction error (Error, %).

The model is also effective in characterizing the drying process of the soil on dry days (July 10, 11, and 14). The model also successfully simulates the increase in moisture following the rain on July 12. The prediction error does not exceed 3.1% for the five days under consideration. This shows that the model is highly accurate in both arid and humid environments.

VII. Discussion and Limitations

The limitations of this research can be better described as definitional. The efficacy of PhyG-Net can be largely attributed to two main factors. Firstly, information integration from increasingly varied spatial indices such as NDVI, along with rapidly changing temporal information streams such as IoT sensor outputs and weather data, presents a more holistic view of the situation, which would not be possible from either modality alone. Secondly, the use of a physics-informed loss function essentially acts as a high-quality regularization mechanism, which minimizes the probability of the network converging on a statistically rather than physically plausible solution.

A. Computational Complexity and Edge Deployment

The PhyG-Net model has shown superior prediction accuracy. However, this accuracy comes with significant computational costs. The memory consumption of the convLSTM layers is significantly higher compared to regular LSTM networks. The empirical analysis of the model has been carried out with regular GPU (NVIDIA Tesla T4) settings. The average time step of 12 milliseconds has been achieved, which is within tolerable limits for cloud computing. However, in edge devices such as Raspberry Pi or ESP32 microcontrollers, model compression is required to facilitate real-time inference.

B. Limitations

There are a number of limitations that need to be acknowledged in this framework.

- The discretization of physical processes adopted by the current model assumes first-order discretization, which implies that the effects of runoff (RO) and deep percolation (D) are considered negligible. Although this might be true in flat terrain, this assumption might not be valid in terrains where steep gradient slopes are present or when heterogeneity is high in the soil layer.
- The linear interpolation of satellite data might miss out on sudden events, such as pest infestations or flood occurrence, which might take place during the interval between satellite observations, which is every five days.
- The weather API is considered an external factor. If there is poor predictive performance of forecasted variables, this might be reflected in the model and might not provide optimal irrigation suggestions.

VIII. Conclusion and Future Work

This manuscript presents a strong and new intelligent irrigation scheduling system based on PhyG-Net. Through the integral fusion of Internet-of-Things data, weather data, and satellite data in a physics-guided deep learning framework, the proposed system can effectively overcome the important limitations of spatial data insufficiency and physical incompatibility existing in previous methods. The developed model achieves strong predictive performance ($R^2 = 0.94$) and meets hydrological constraints, providing useful irrigation scheduling information for agronomists.

The future research will be focused on three main directions:

1) Integration with automated solenoid actuators and variable-rate irrigation systems to create a self-sustaining closed-loop agricultural control system.
2) Comprehensive evaluation and optimization on lightweight edge AI platforms (such as Raspberry Pi and NVIDIA Jetson) to support offline inference in geographically isolated regions.
3) Extension of the physics model to handle different soil types and varying runoff conditions, with the possibility of using Graph Neural Networks to model the topological complexity of water flow in agricultural fields.

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